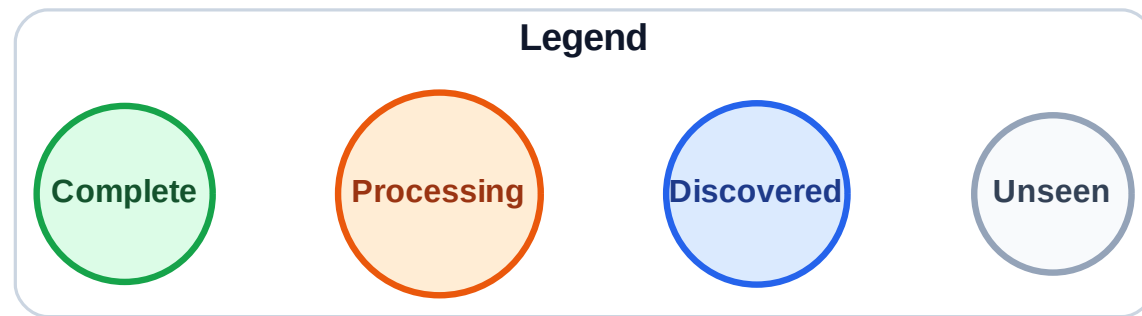


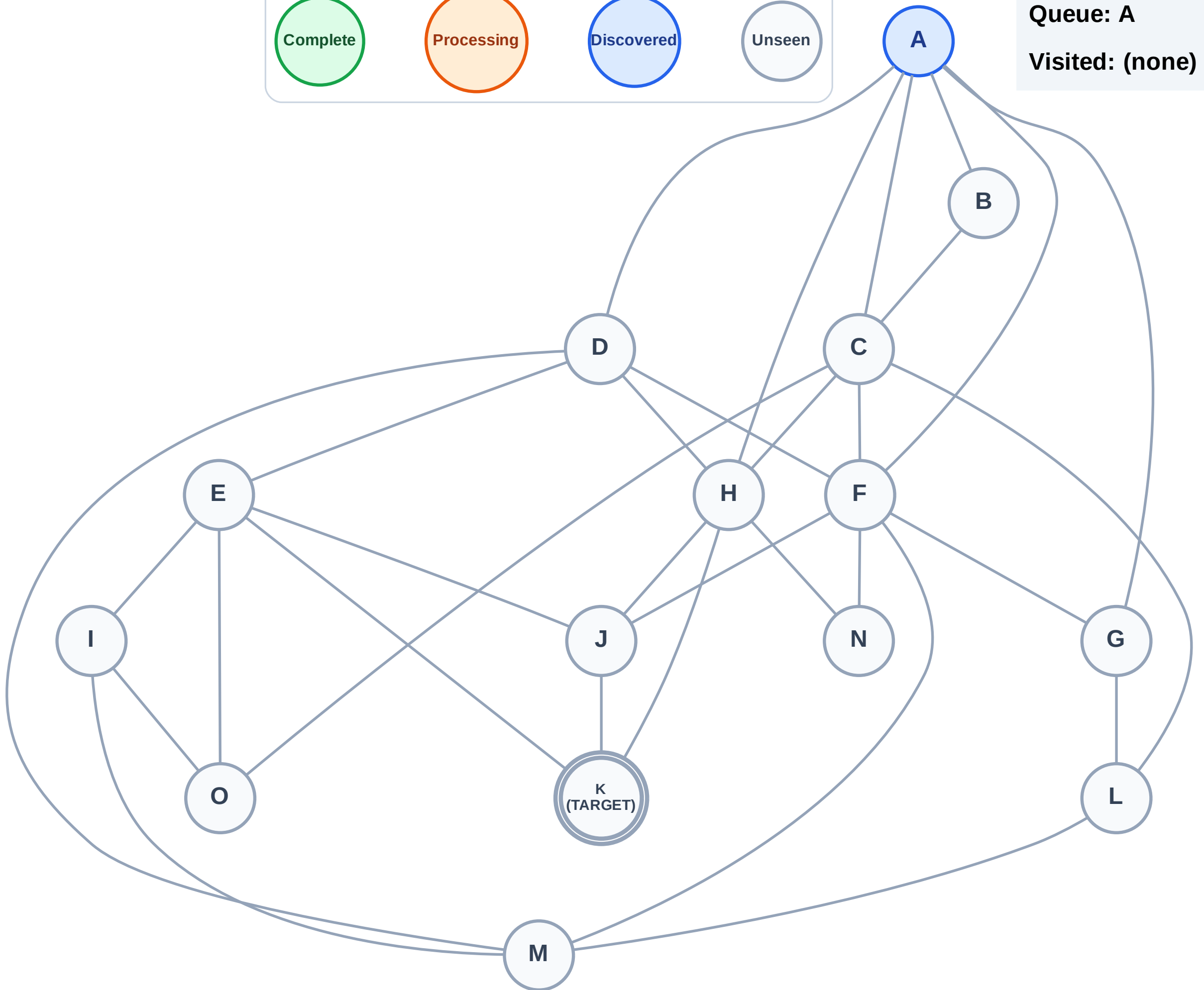
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A

Visited: (none)



BFS Traversal

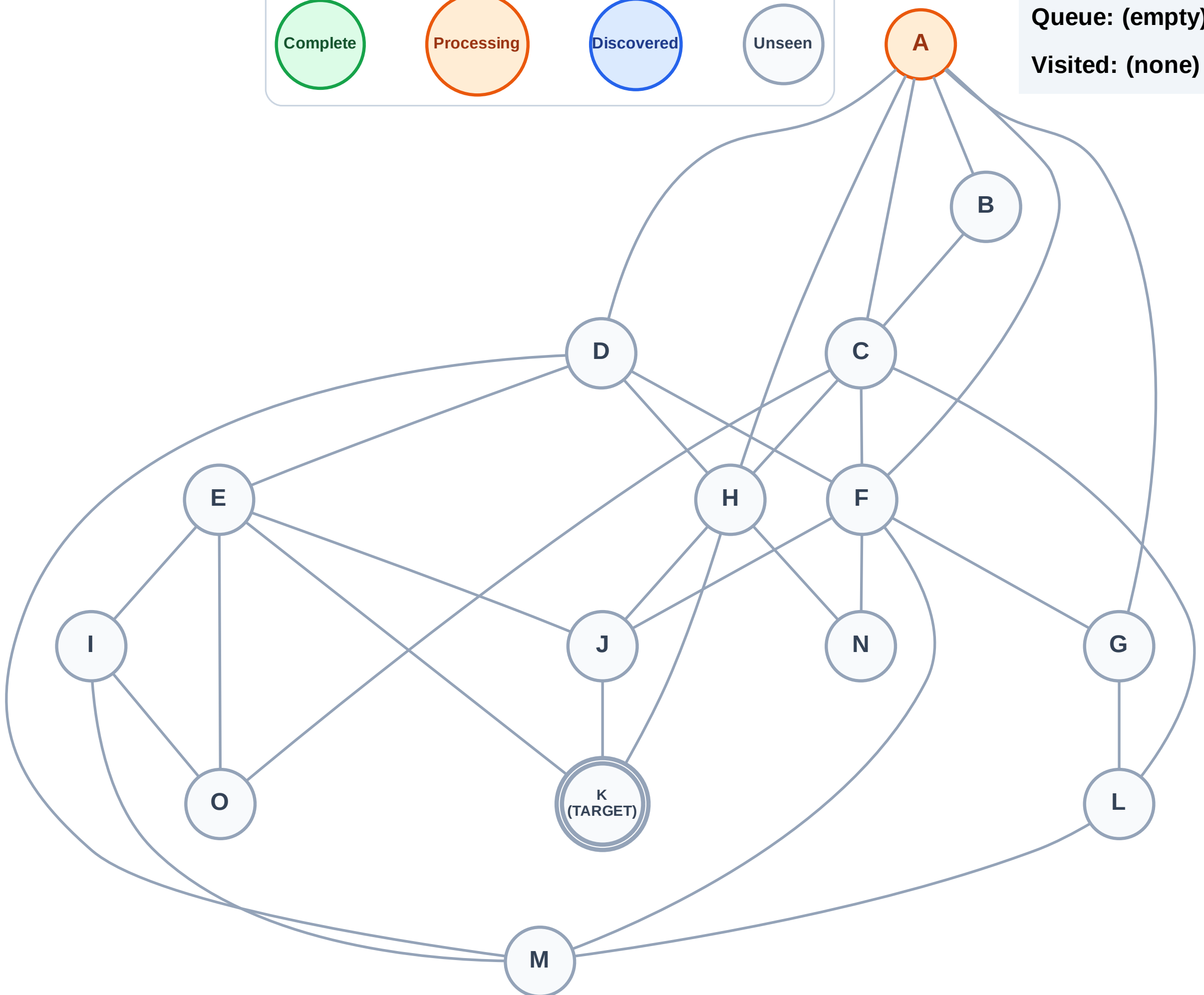
Step 2: Process node A

Legend



Queue: (empty)

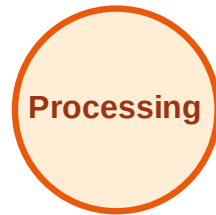
Visited: (none)



BFS Traversal

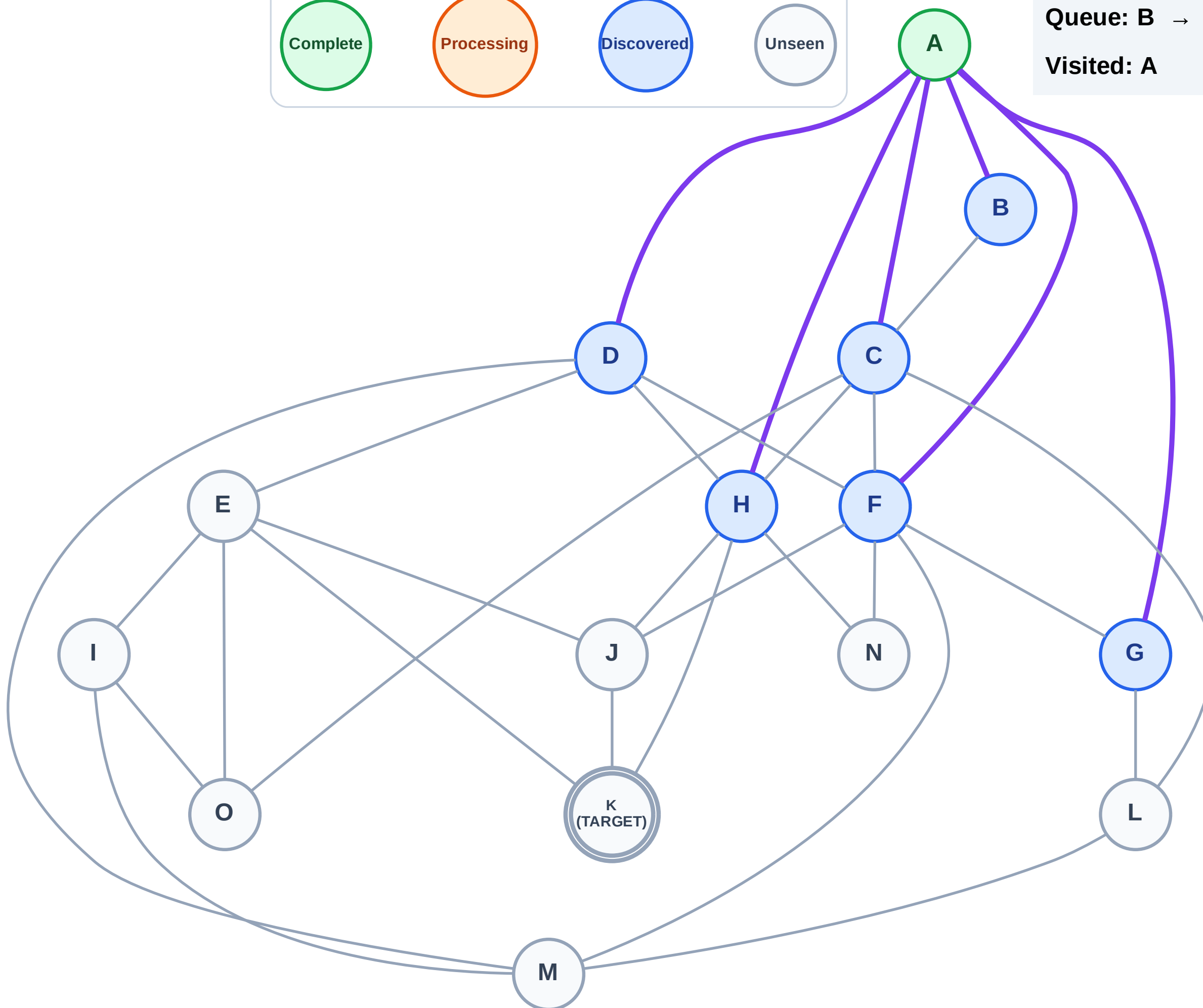
Step 3: Discover B, C, D, F, G, H from A

Legend



Queue: B → C → D → F → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

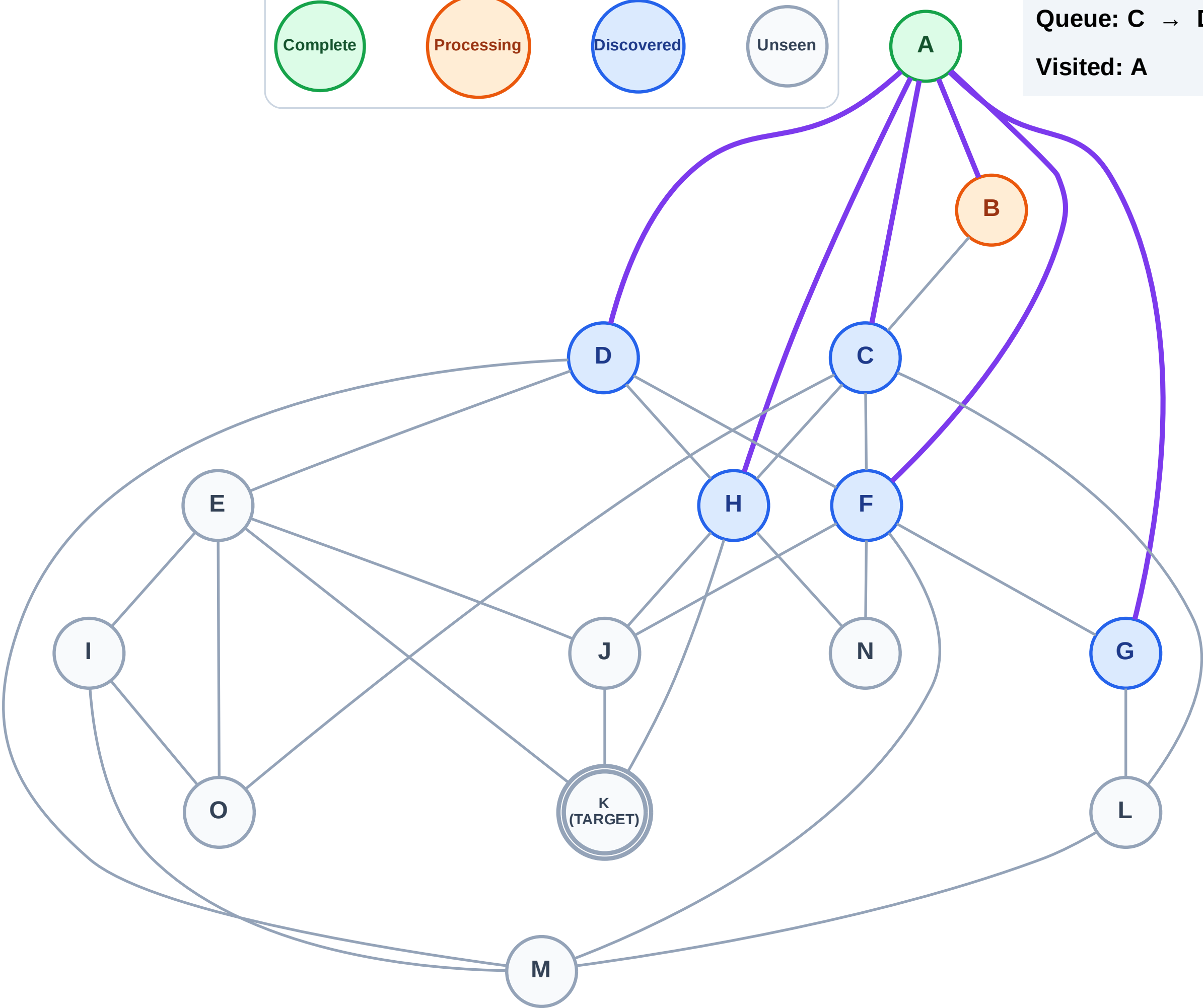
Processing

Discovered

Unseen

Queue: C → D → F → G → H

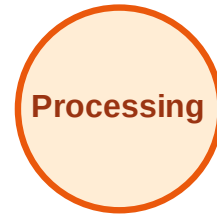
Visited: A



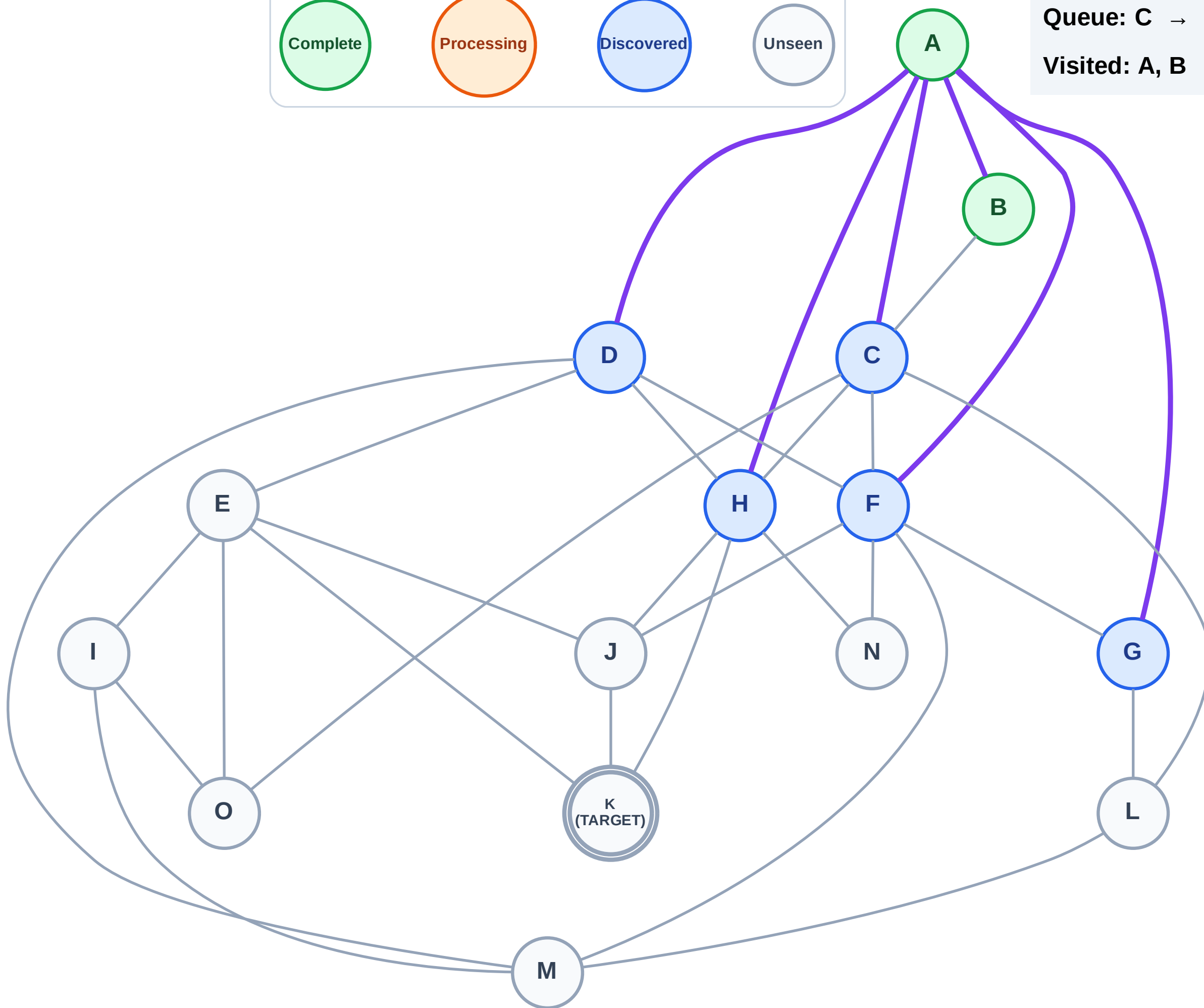
BFS Traversal

Step 5: No new neighbors from B

Legend



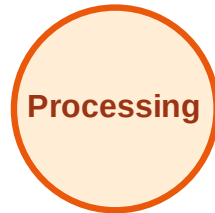
Queue: C → D → F → G → H
Visited: A, B



BFS Traversal

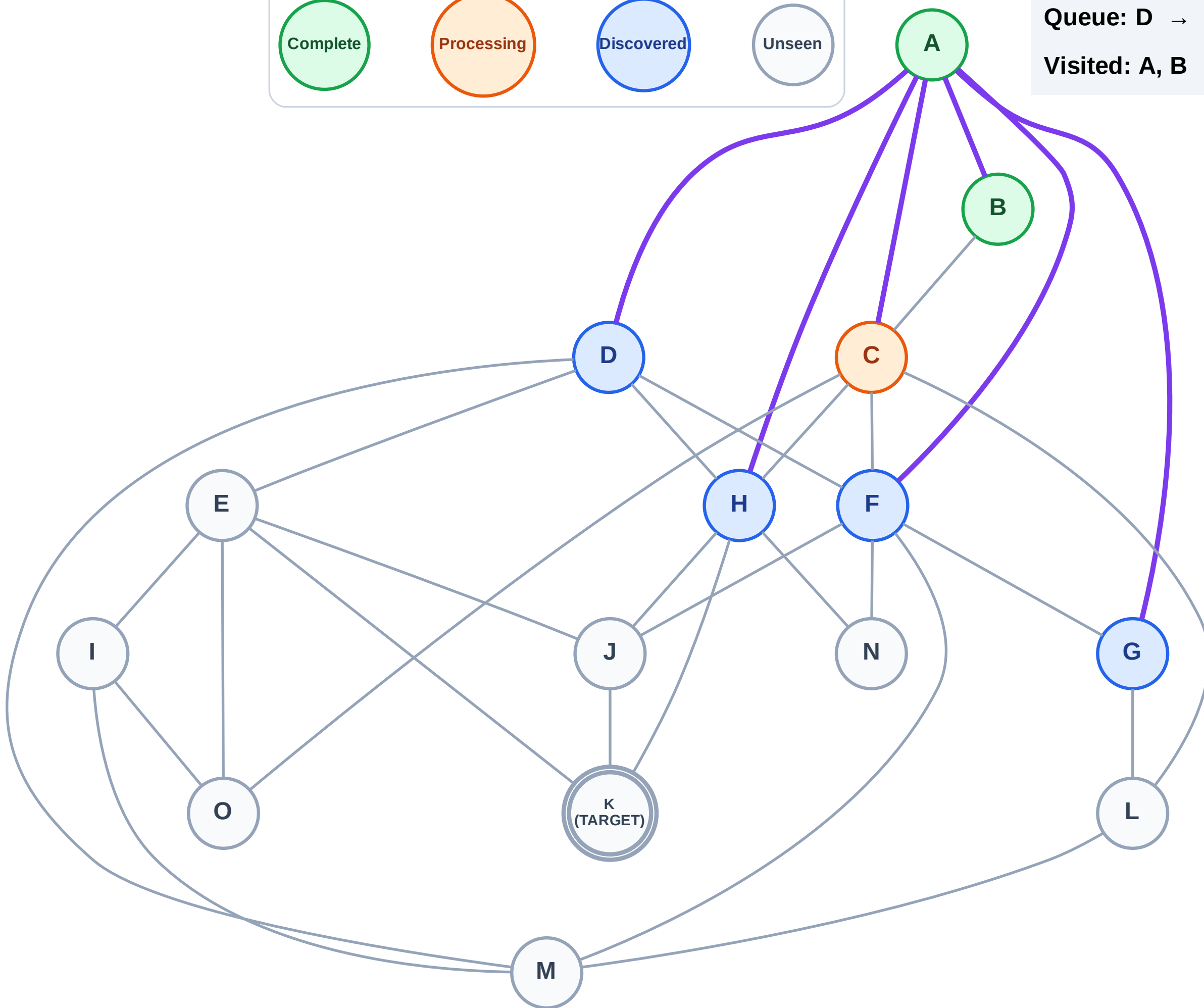
Step 6: Process node C

Legend



Queue: D → F → G → H

Visited: A, B



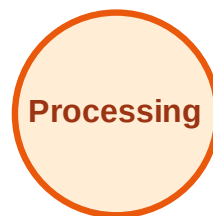
BFS Traversal

Step 7: Discover L, O from C

Legend



Complete



Processing



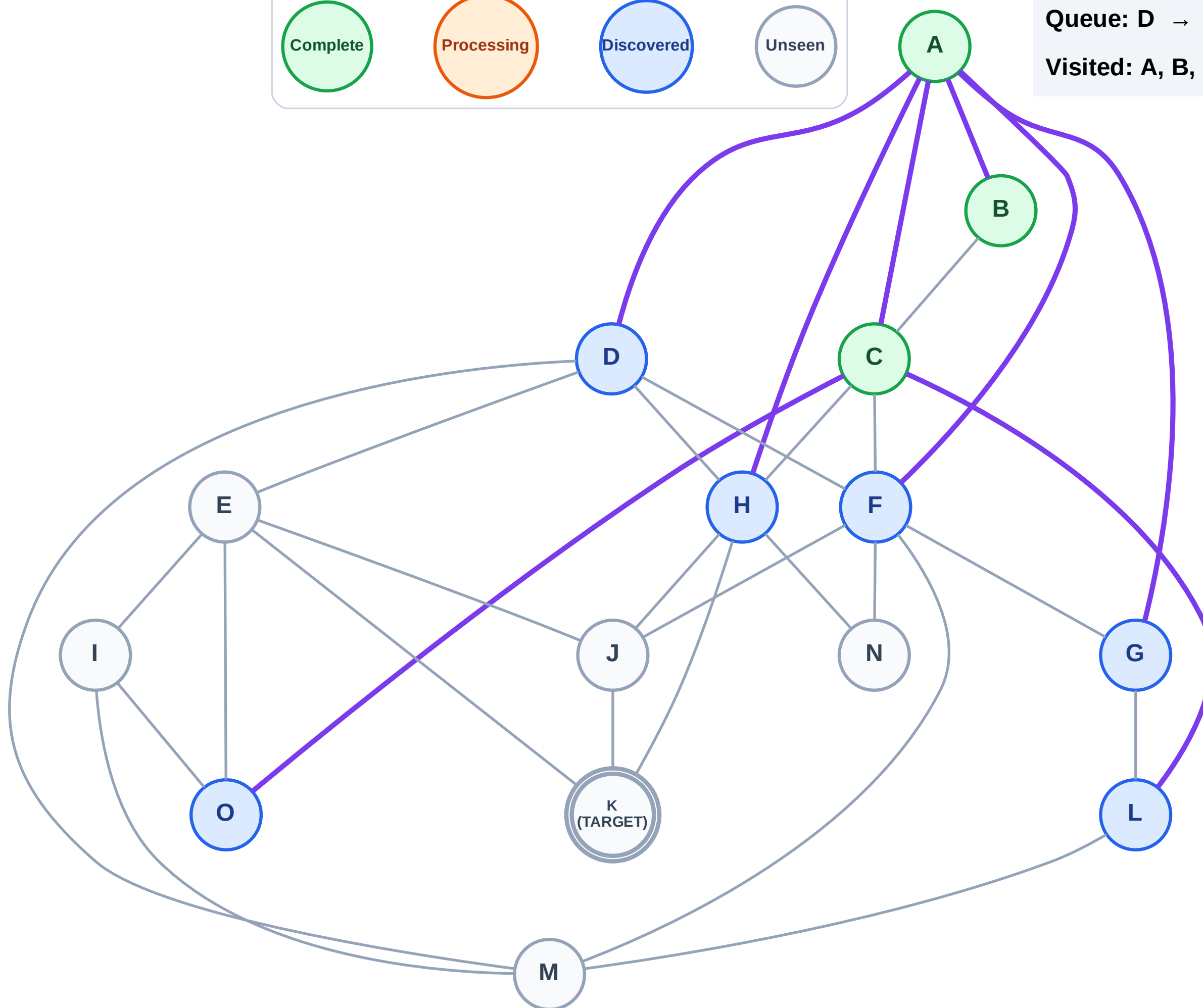
Discovered



Unseen

Queue: D → F → G → H → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

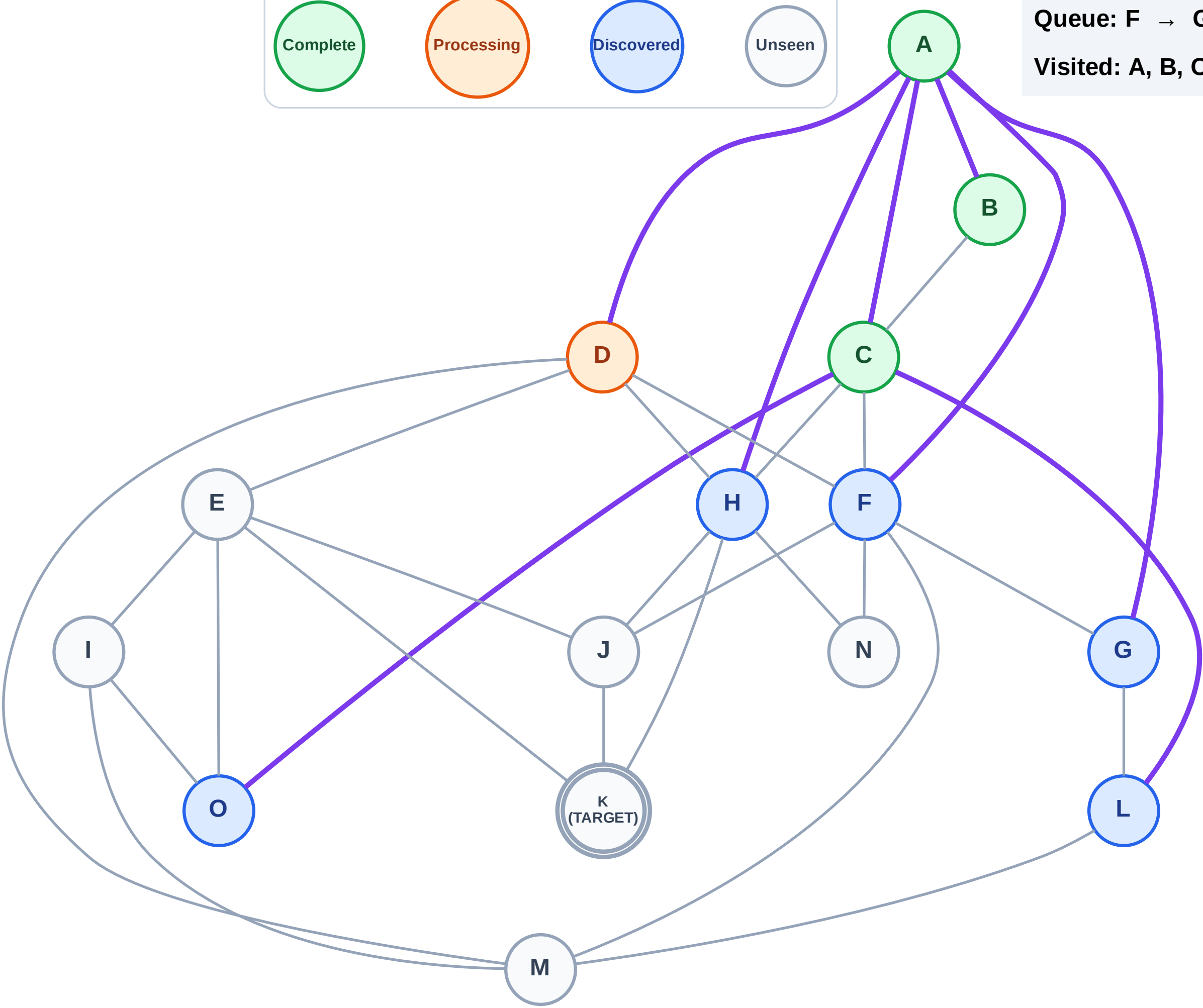
Processing

Discovered

Unseen

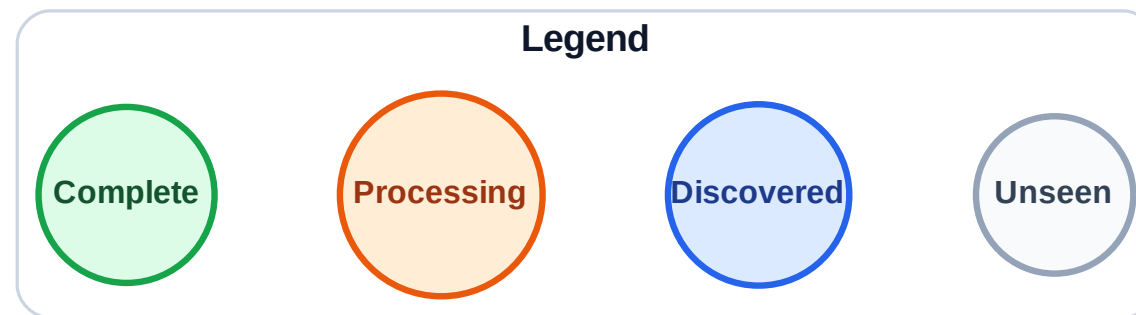
Queue: F → G → H → L → O

Visited: A, B, C



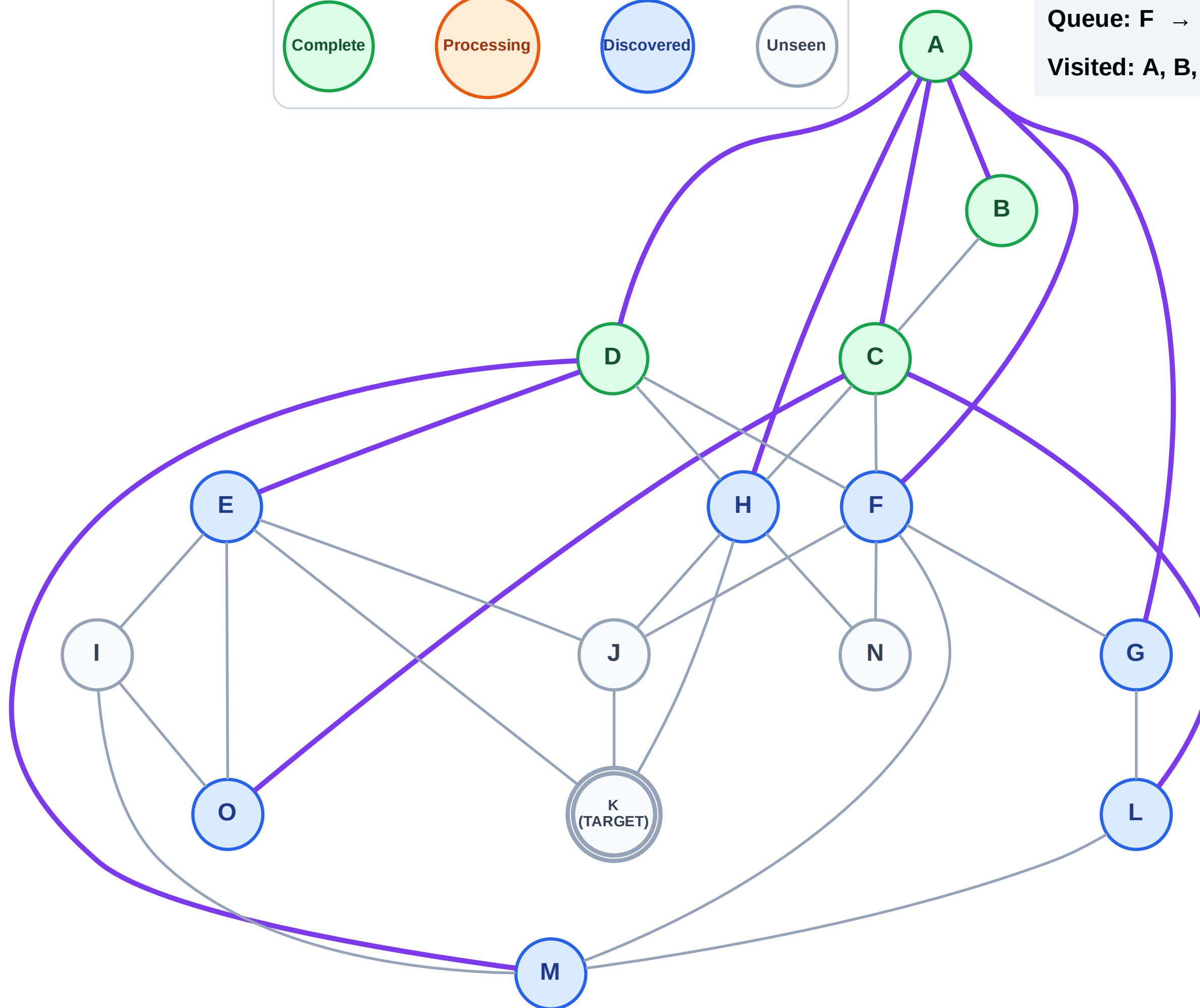
Step 9: Discover E, M from D

Step 9: Discover E, M from D



Queue: F → G → H → L → O → E → M

Visited: A, B, C, D



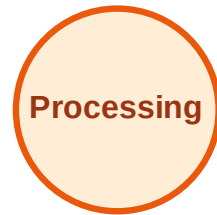
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



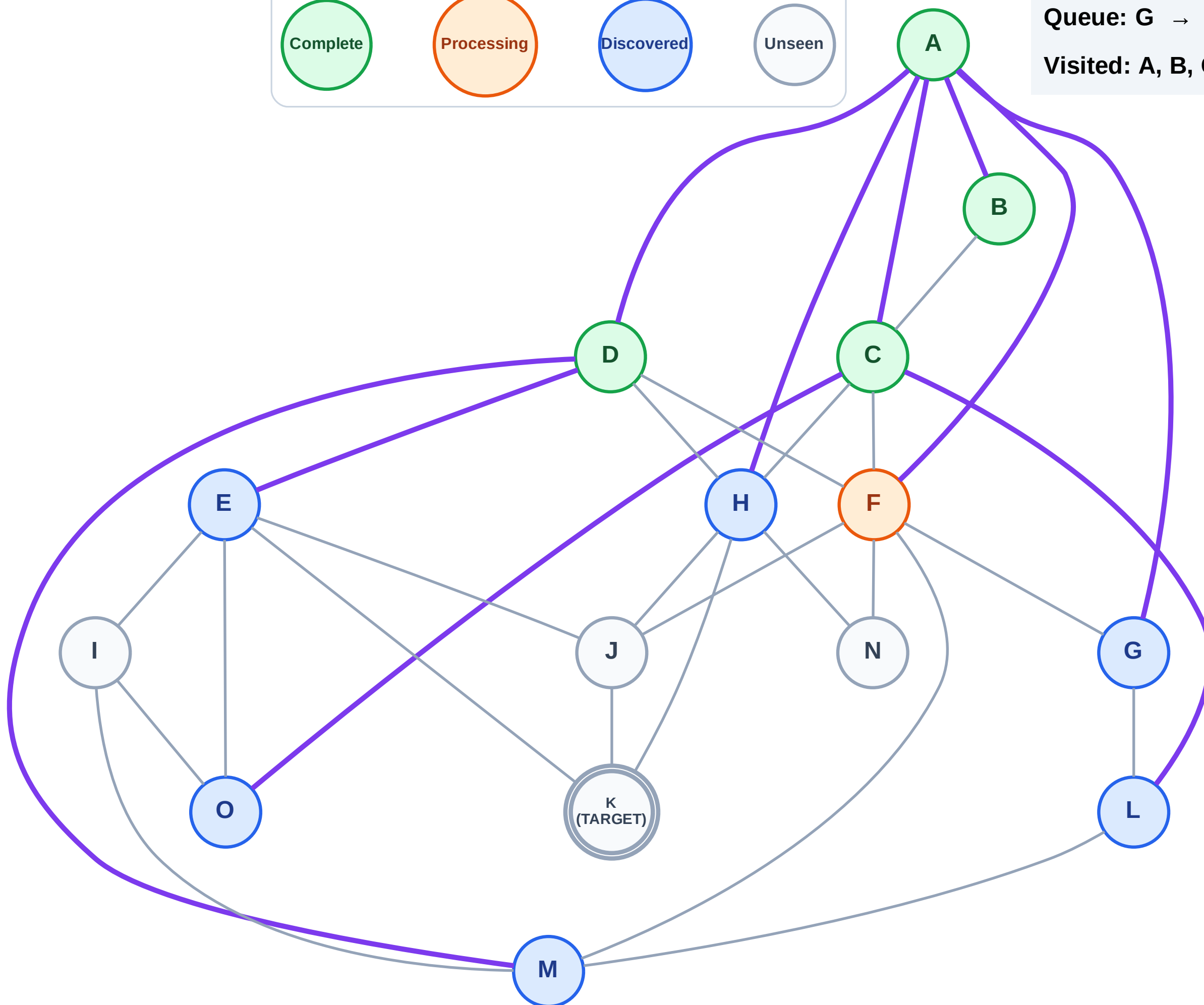
Discovered



Unseen

Queue: G → H → L → O → E → M

Visited: A, B, C, D



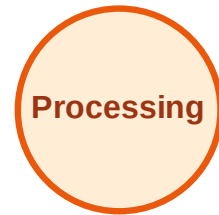
BFS Traversal

Step 11: Discover J, N from F

Legend



Complete



Processing



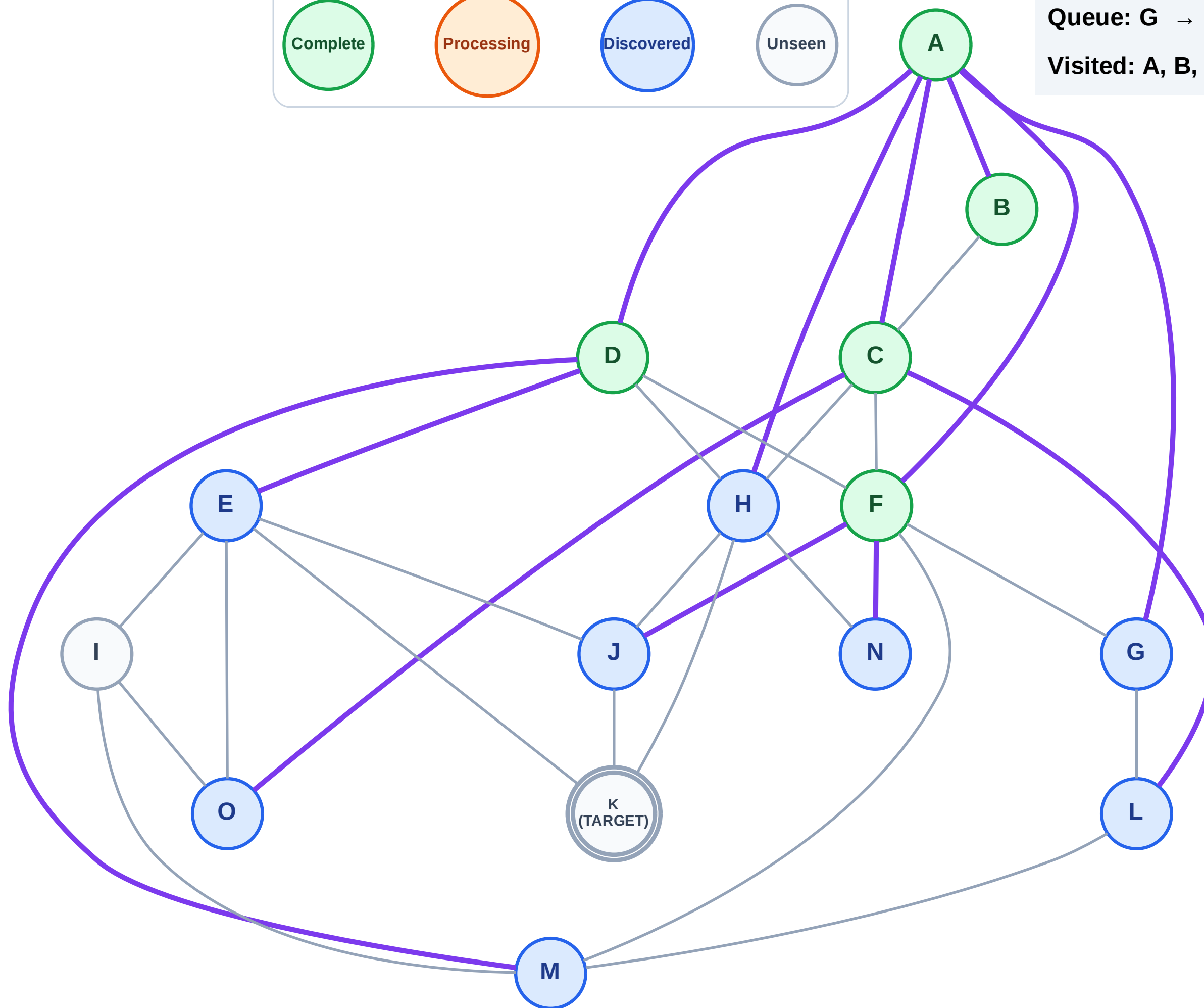
Discovered



Unseen

Queue: G → H → L → O → E → M → J → N

Visited: A, B, C, D, F



BFS Traversal

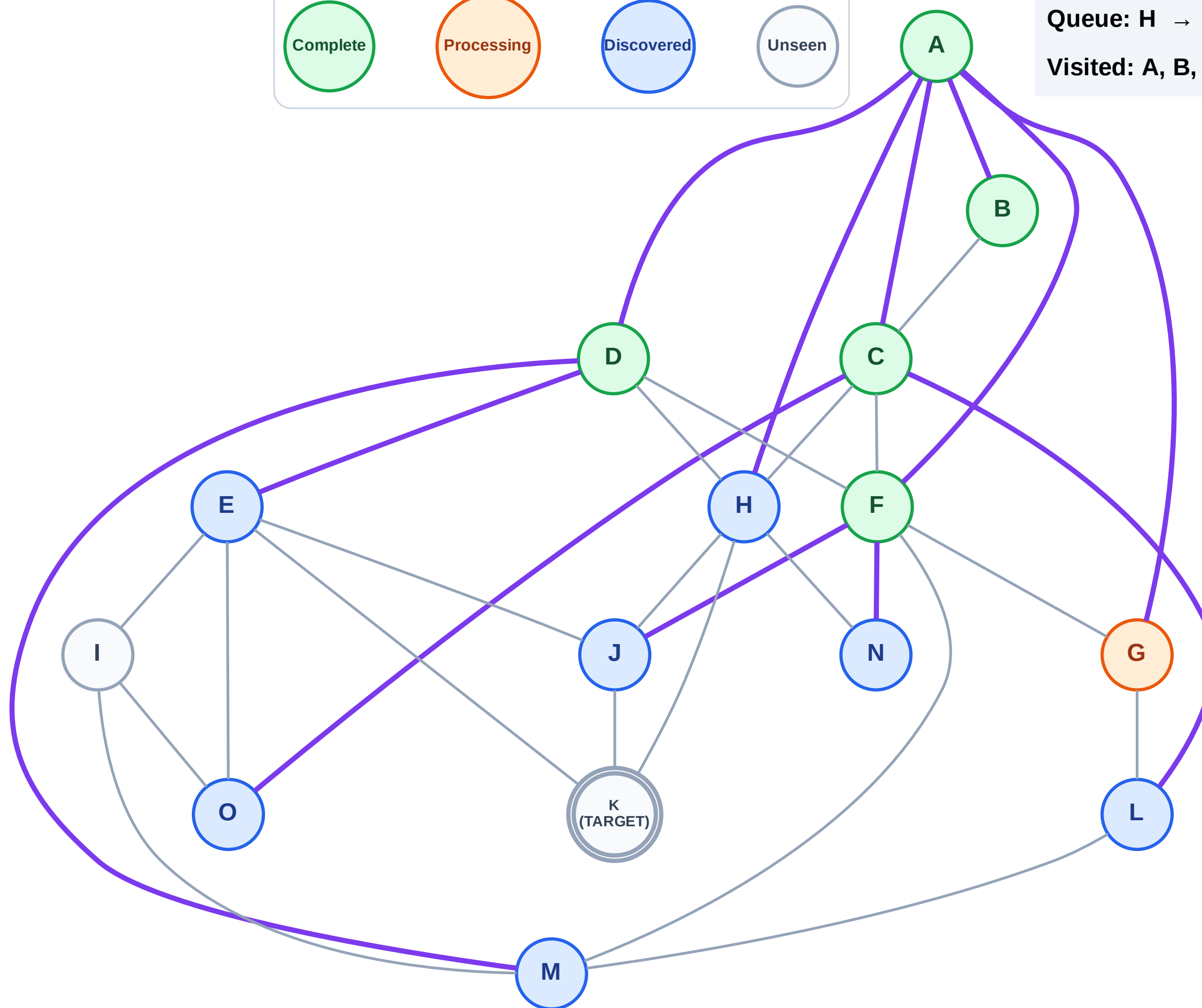
Step 12: Process node G

Legend



Queue: H → L → O → E → M → J → N

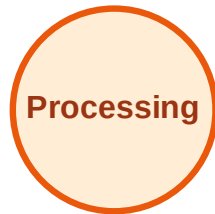
Visited: A, B, C, D, F



BFS Traversal

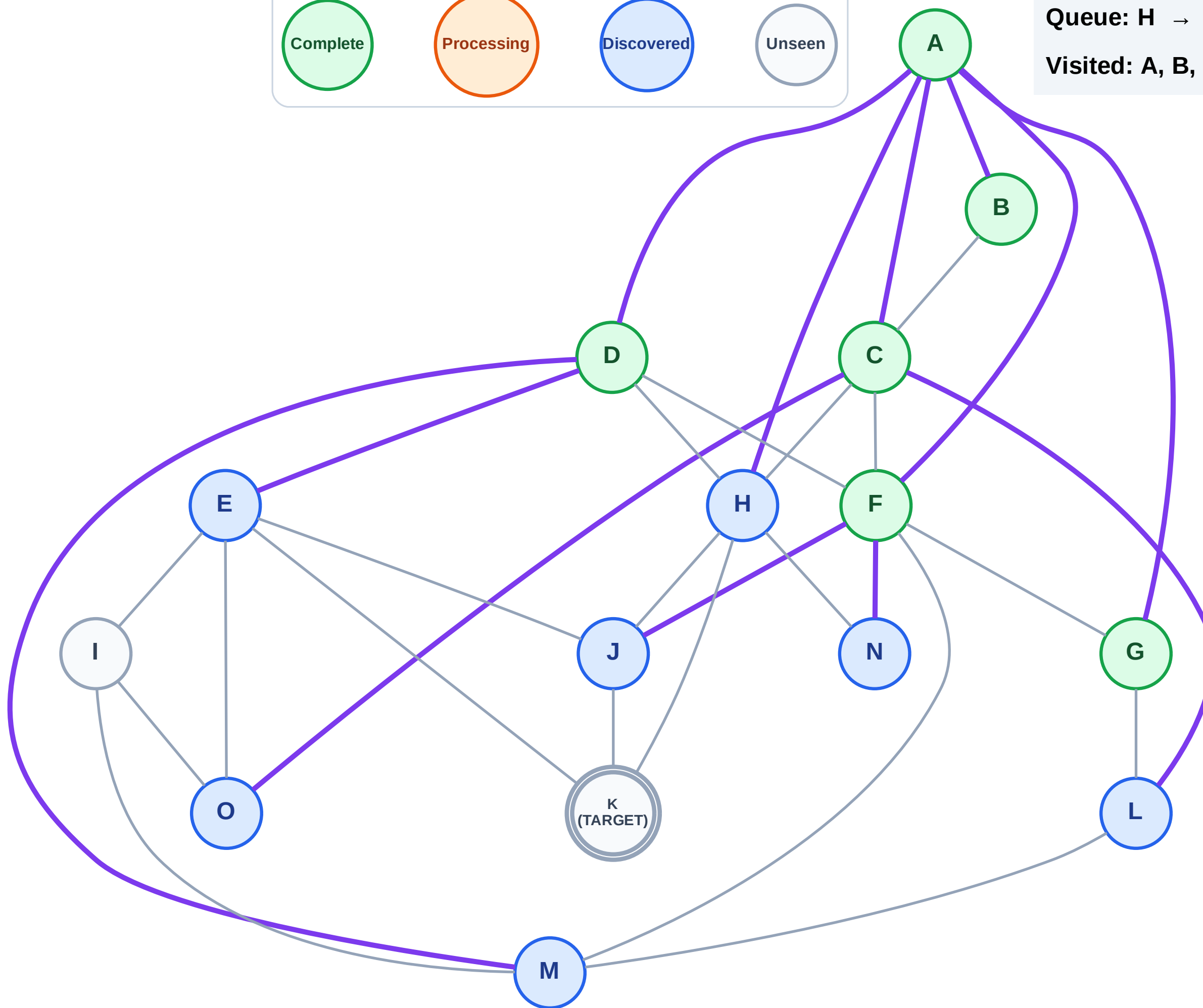
Step 13: No new neighbors from G

Legend



Queue: H → L → O → E → M → J → N

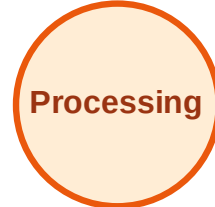
Visited: A, B, C, D, F, G



BFS Traversal

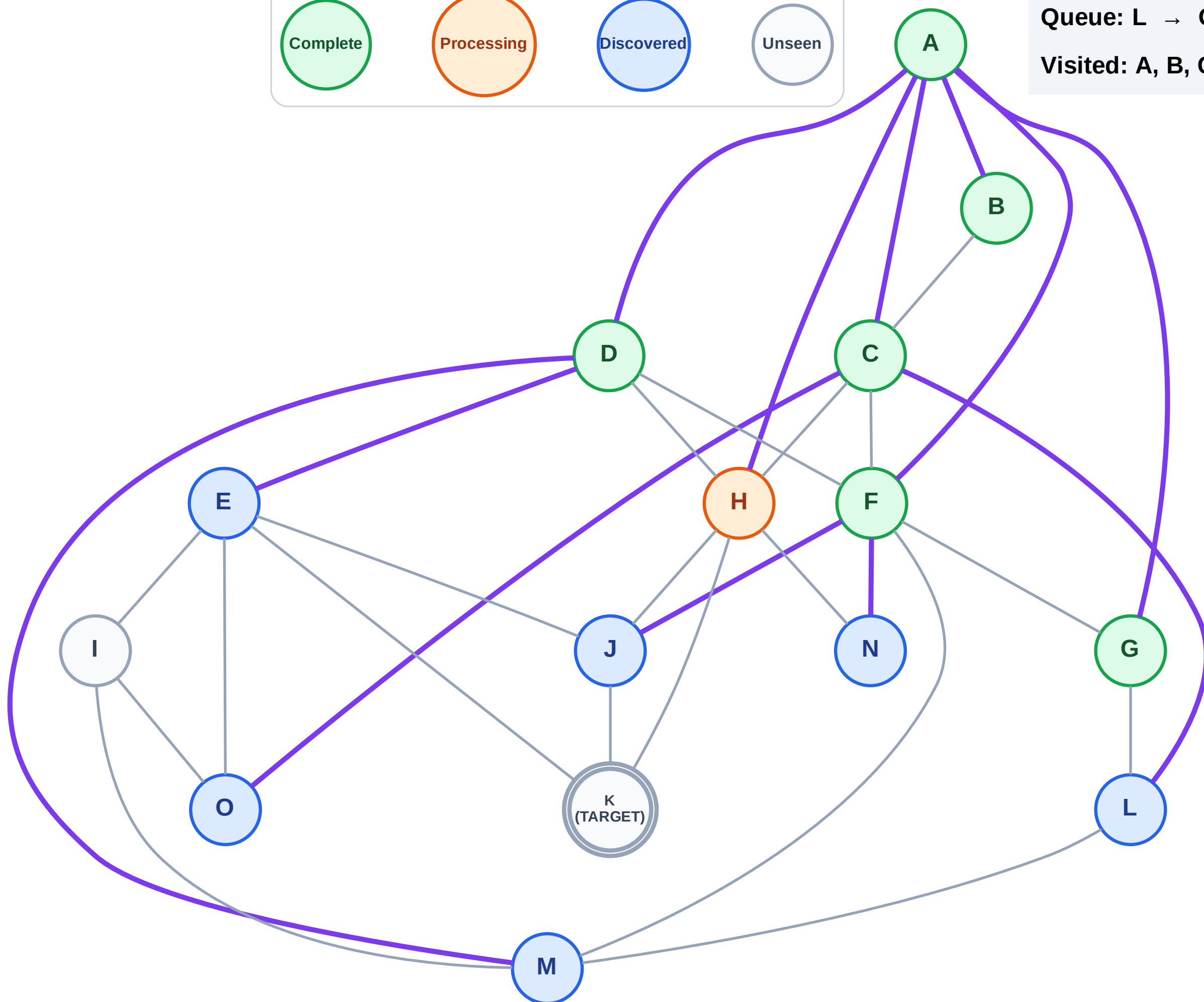
Step 14: Process node H

Legend



Queue: L → O → E → M → J → N

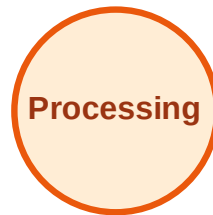
Visited: A, B, C, D, F, G



BFS Traversal

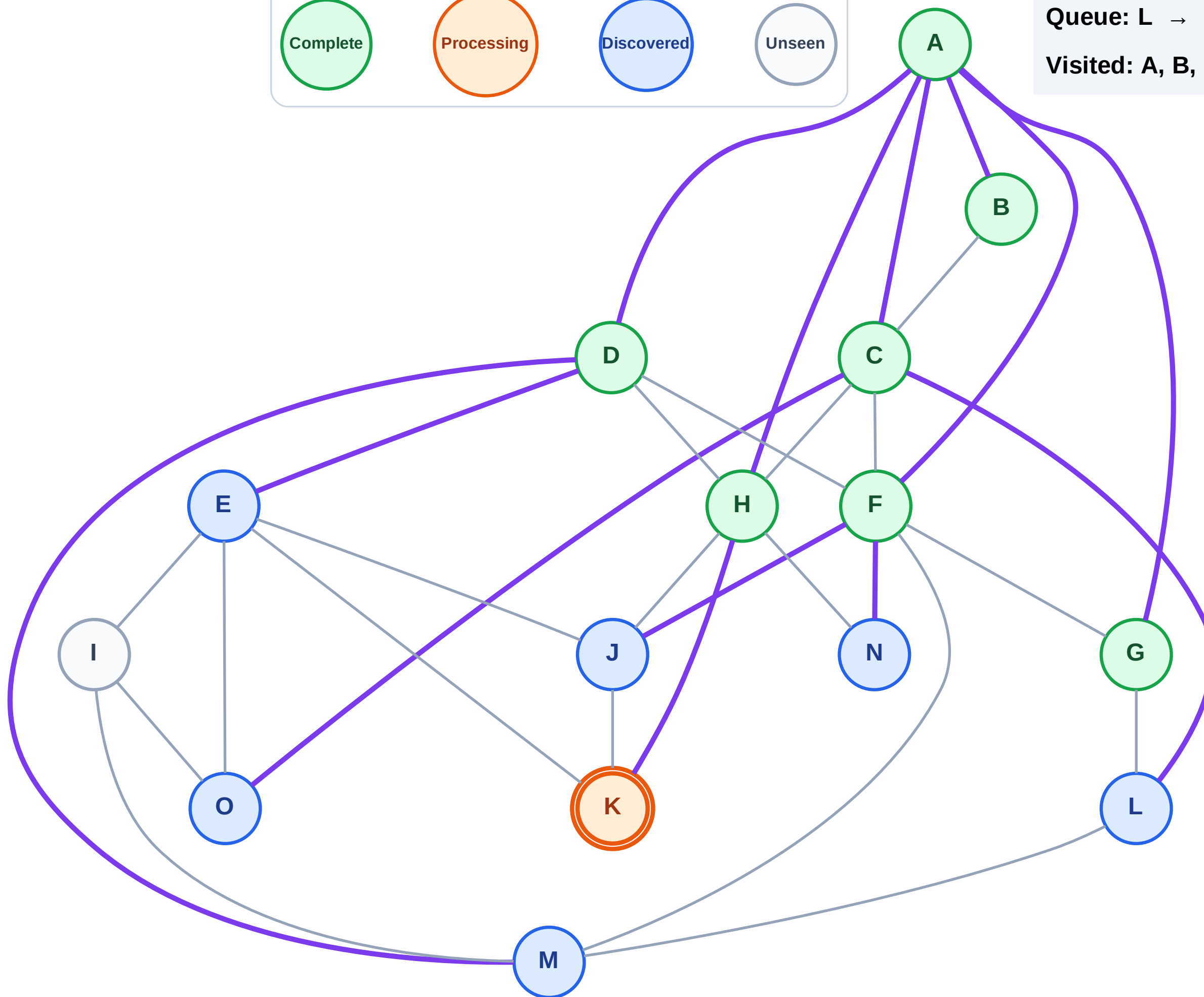
Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → H → K

Legend



Queue: L → O → E → M → J → N

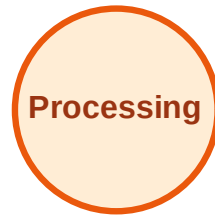
Visited: A, B, C, D, F, G, H, K



DFS Traversal

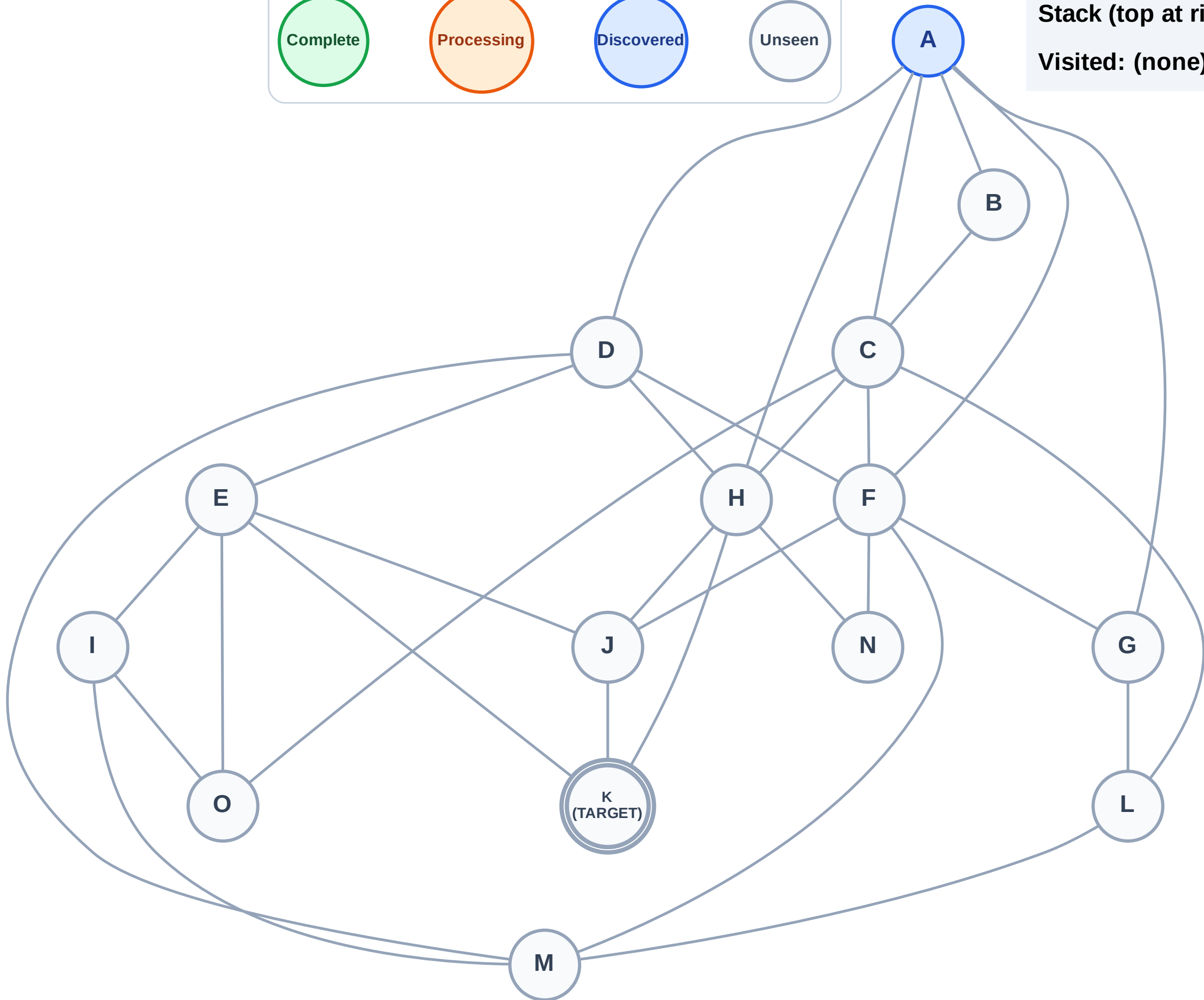
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

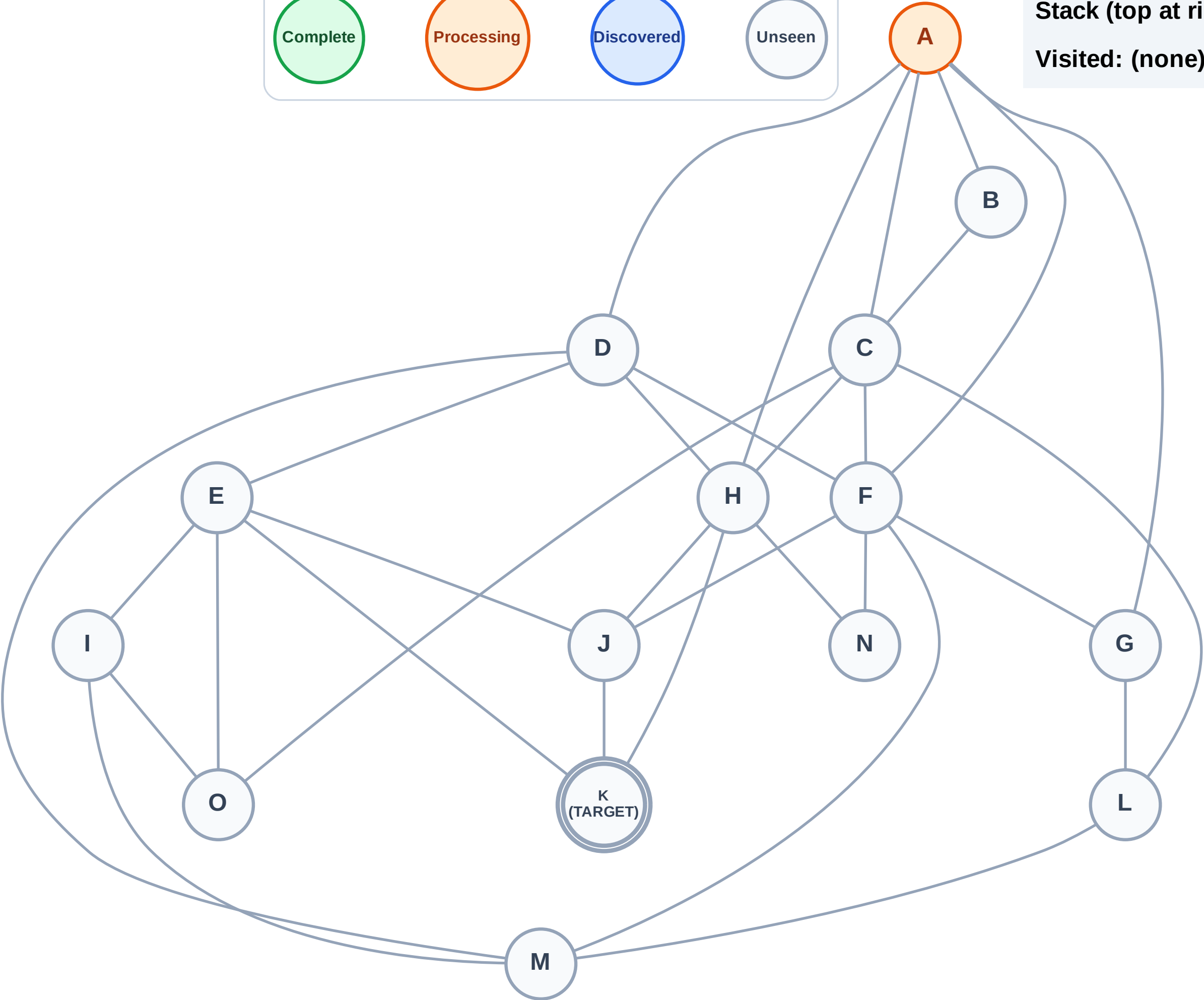
Complete

Processing

Discovered

Unseen

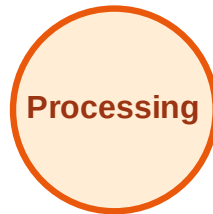
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

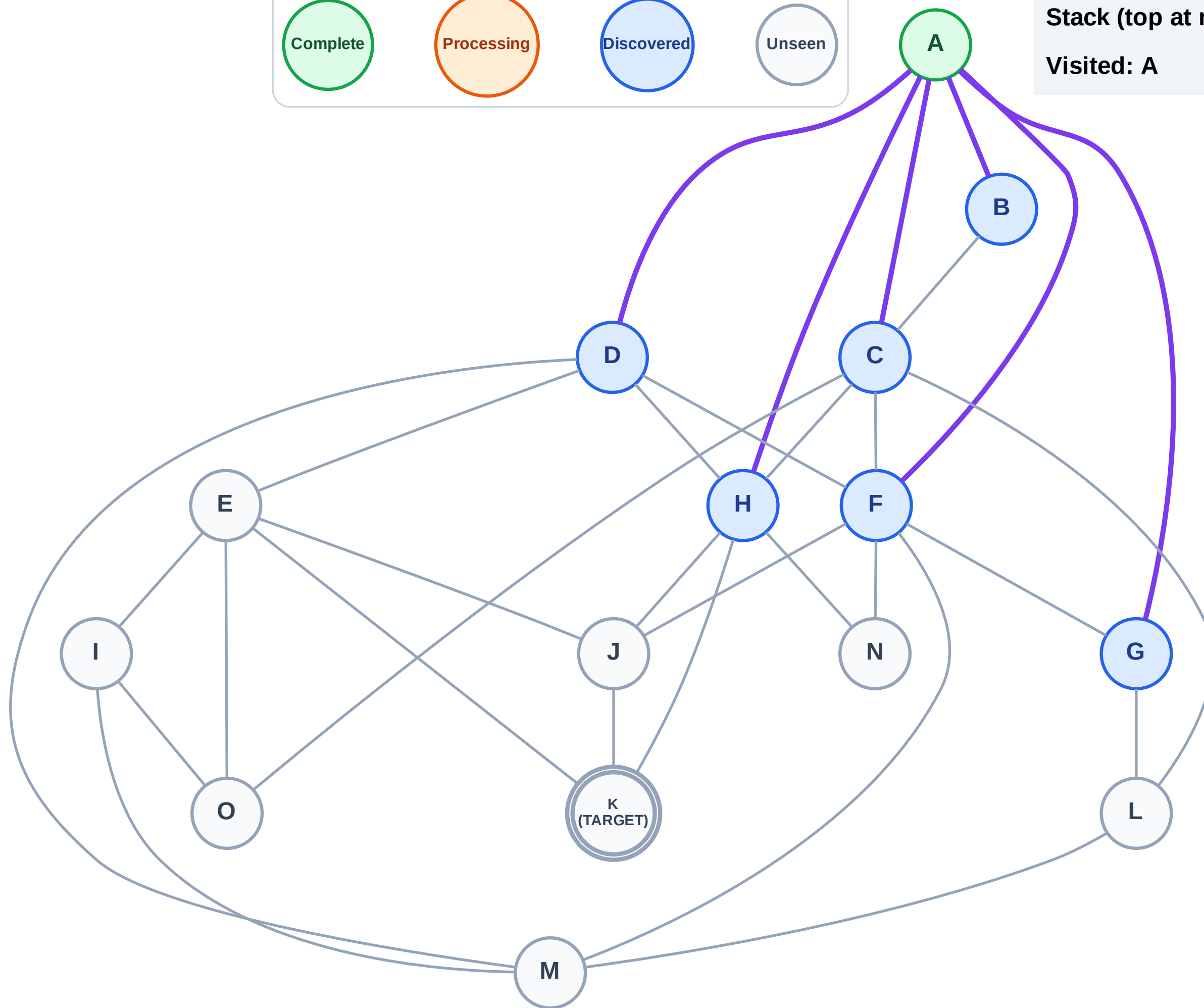
Step 3: Discover H, G, F, D, C, B from A

Legend



Stack (top at right): H | G | F | D | C | B

Visited: A



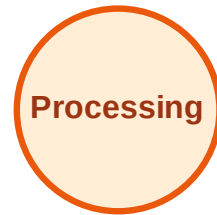
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



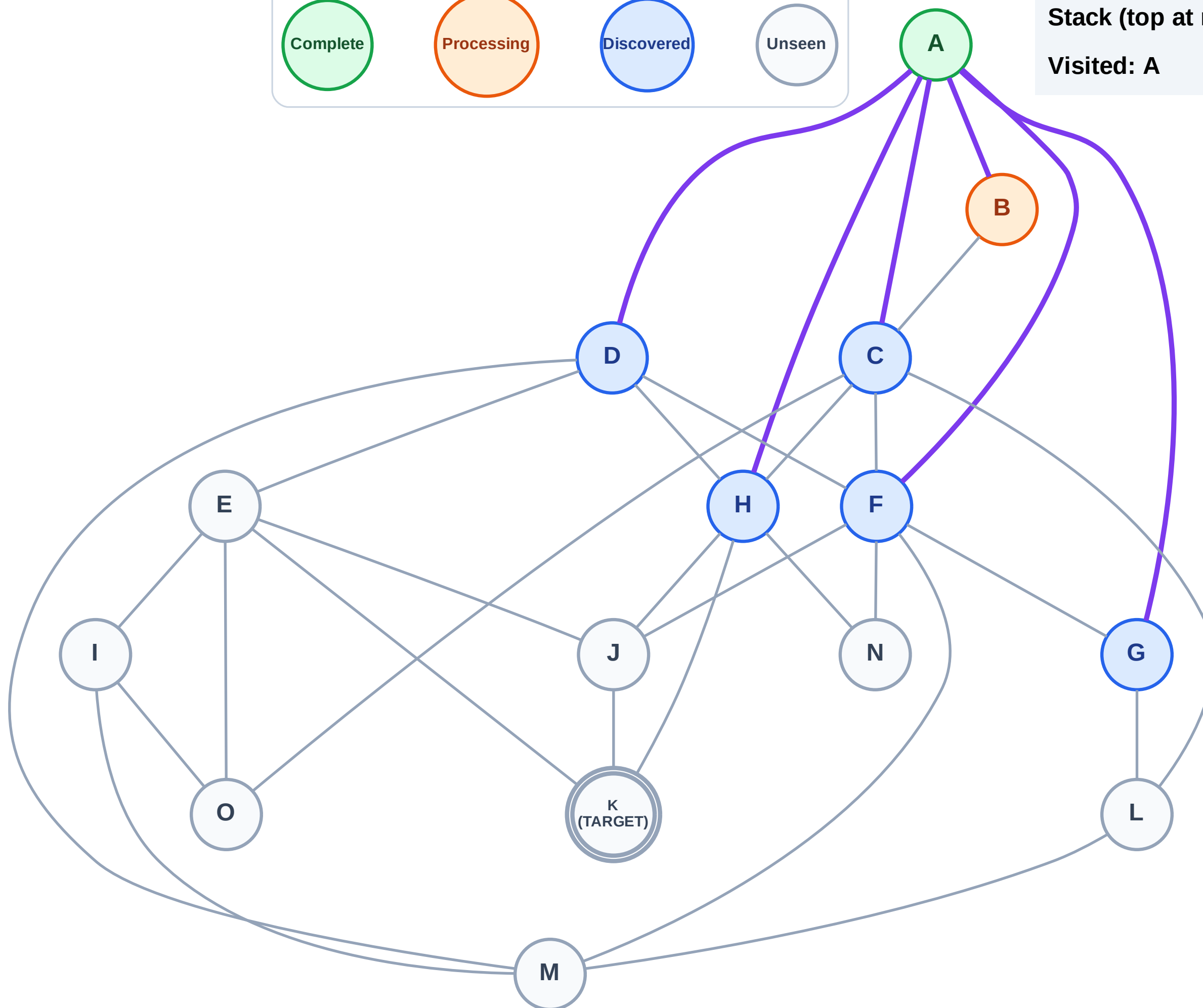
Discovered



Unseen

Stack (top at right): H | G | F | D | C

Visited: A



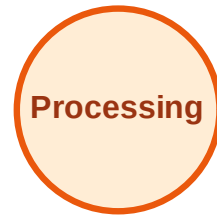
DFS Traversal

Step 5: No new neighbors from B

Legend



Complete



Processing



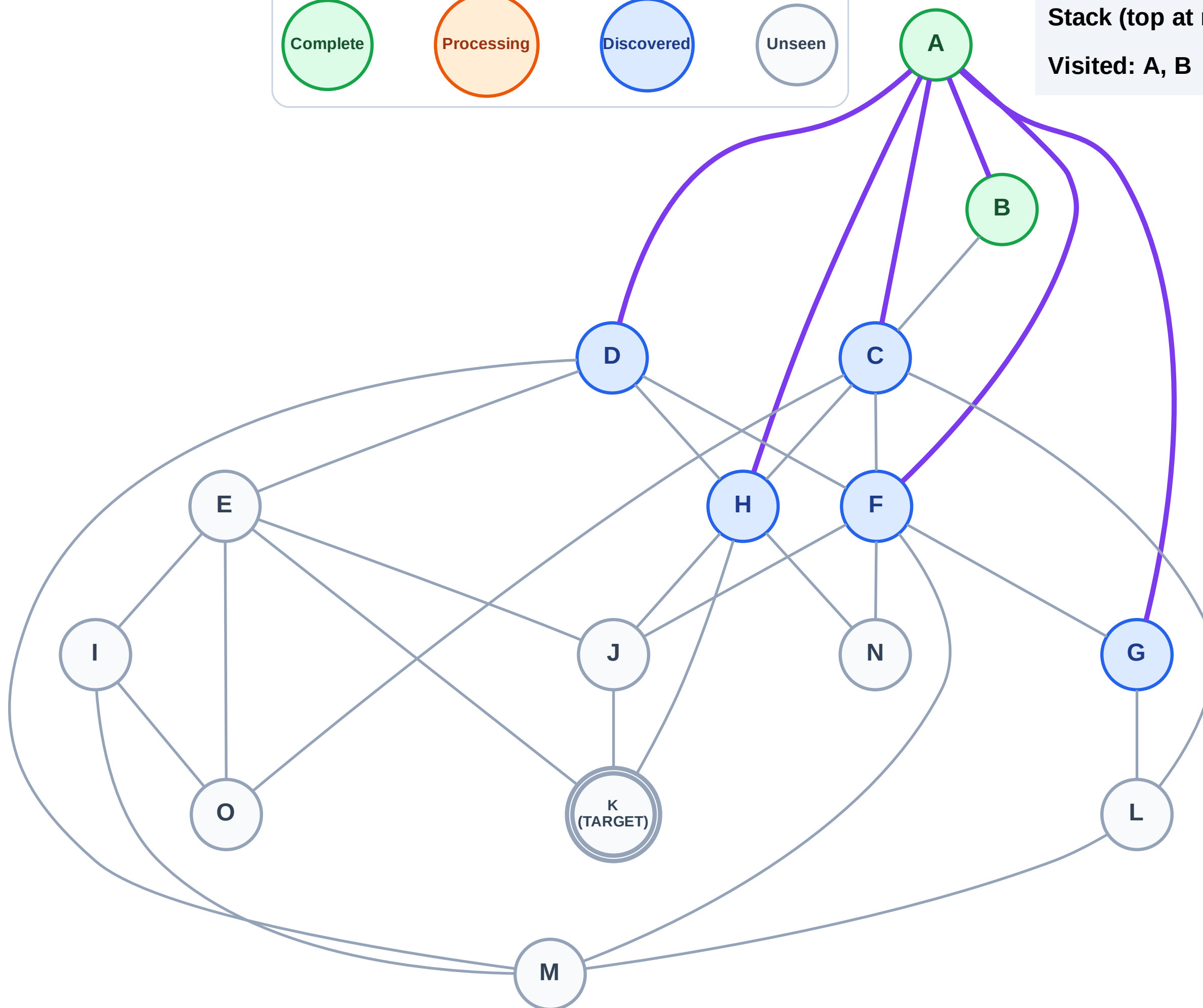
Discovered



Unseen

Stack (top at right): H | G | F | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

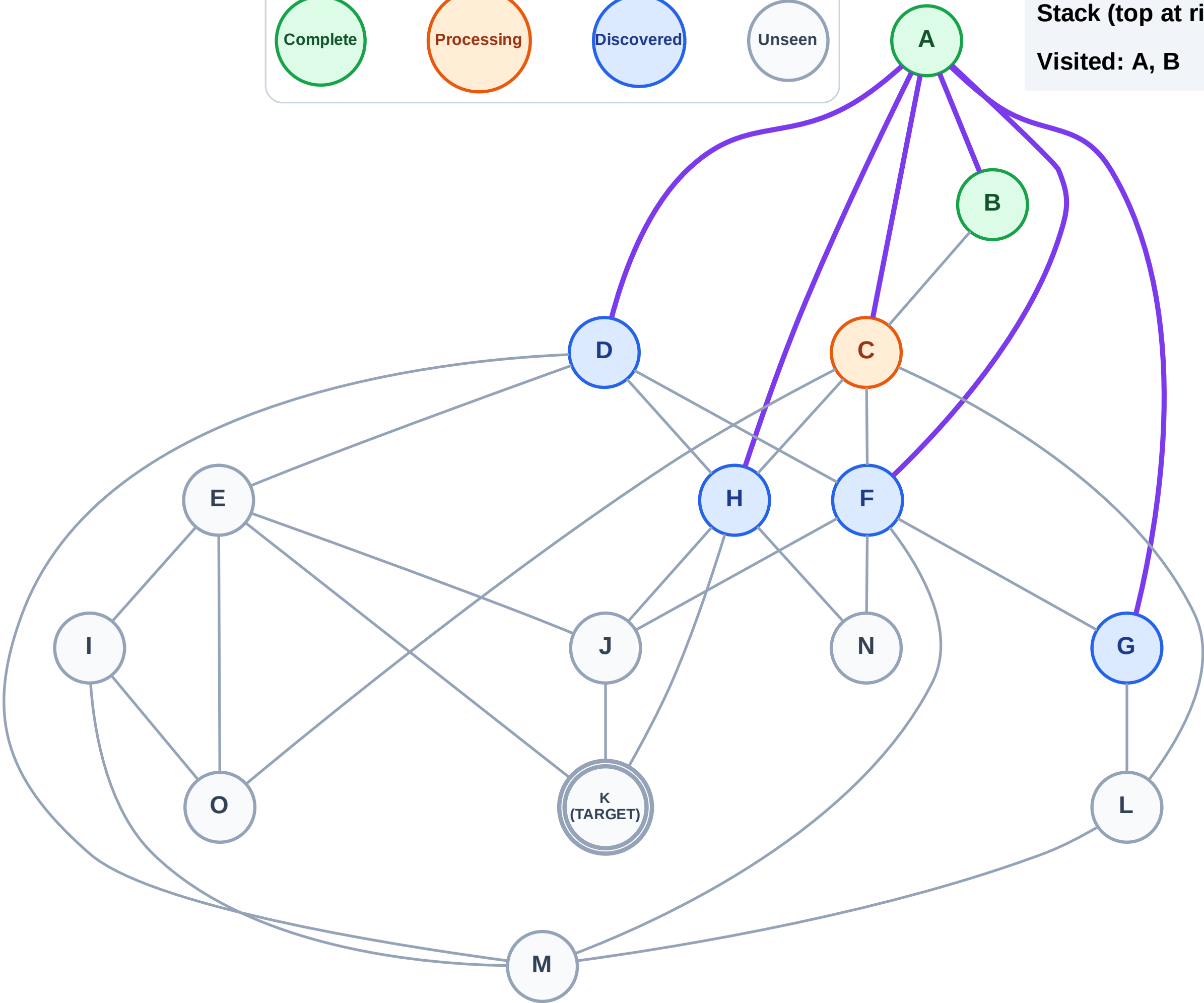
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D

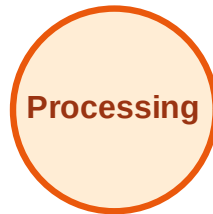
Visited: A, B



DFS Traversal

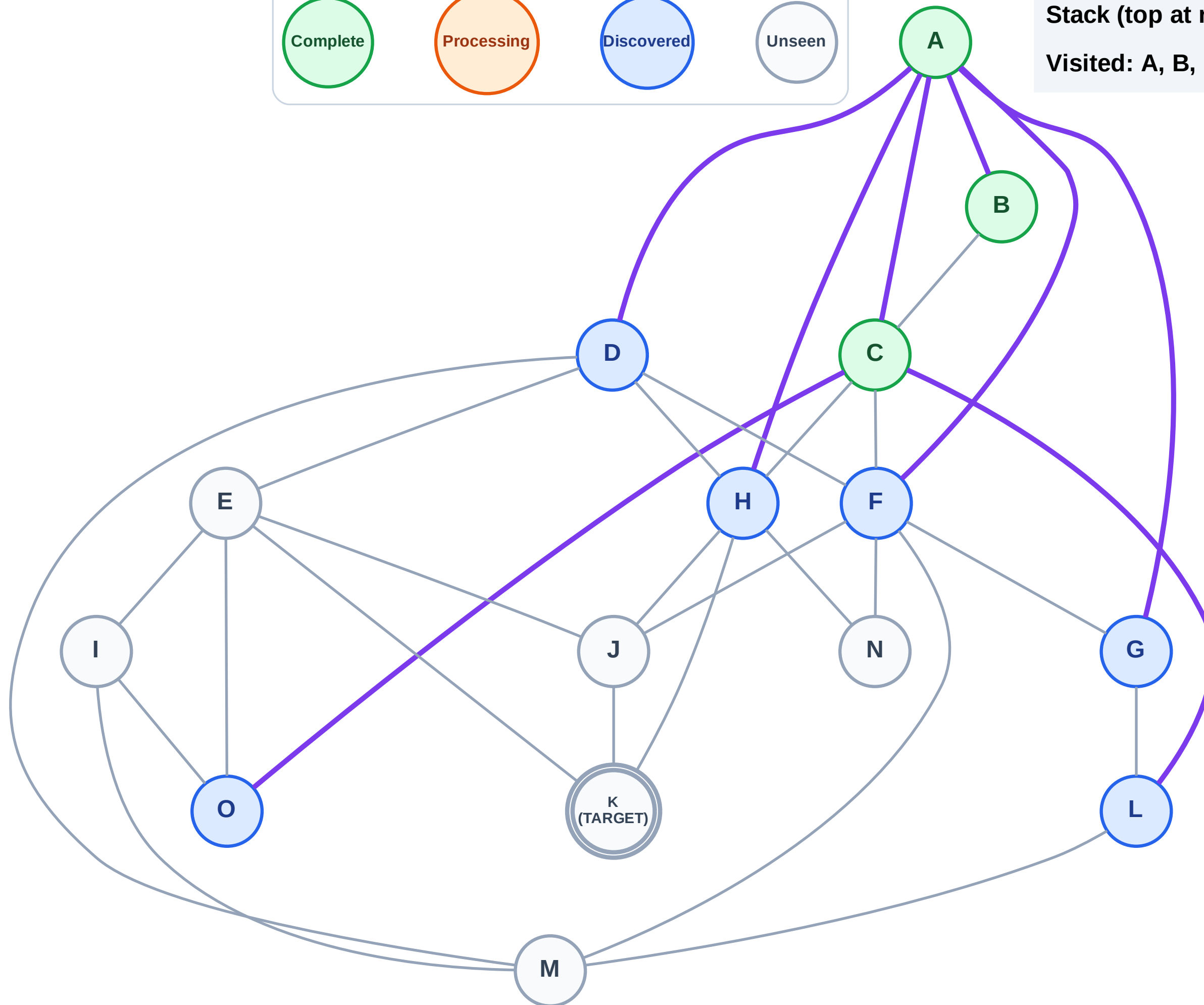
Step 7: Discover O, L from C

Legend



Stack (top at right): H | G | F | D | O | L

Visited: A, B, C



DFS Traversal

Step 8: Process node L

Legend

Complete

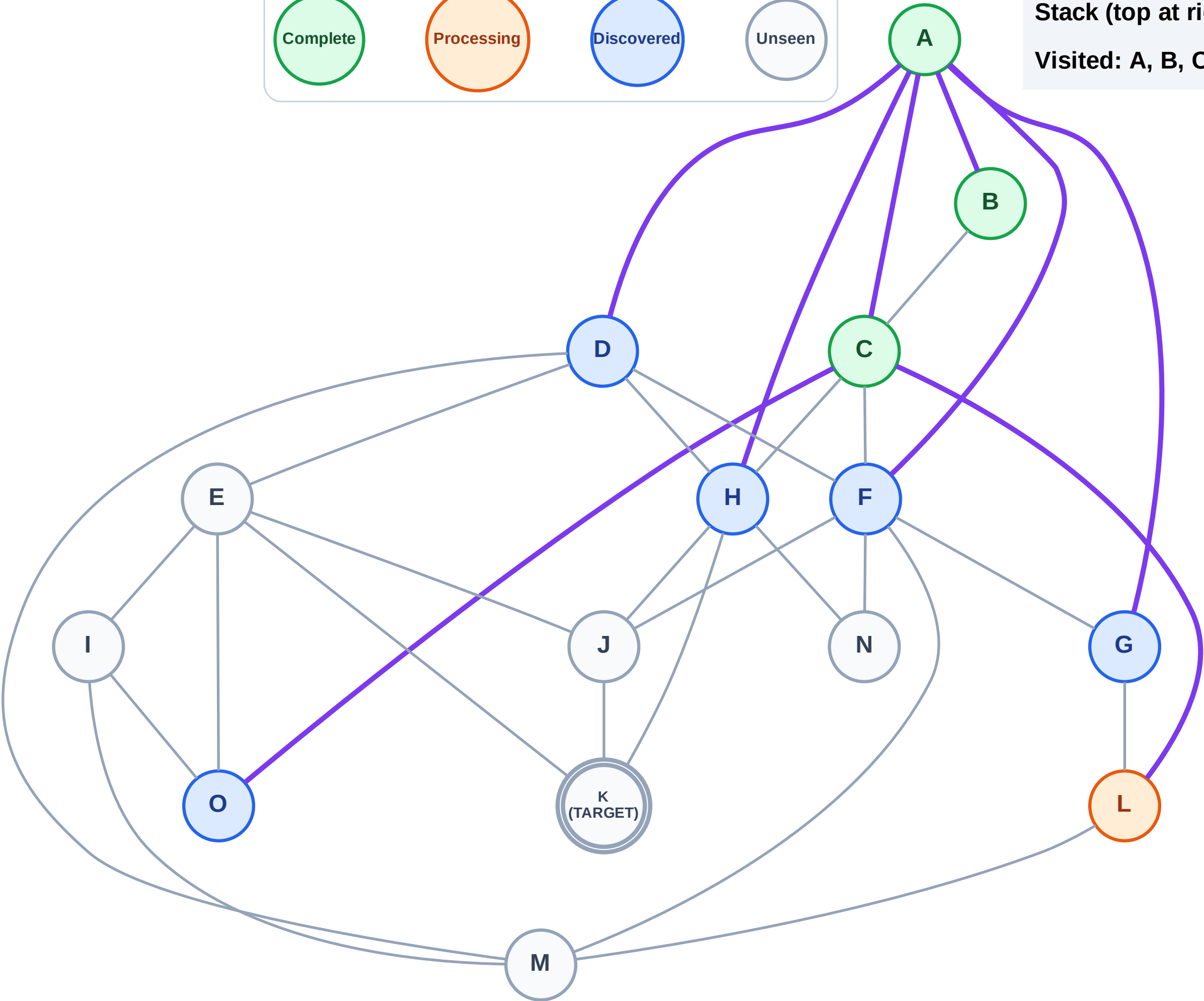
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | O

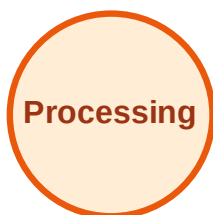
Visited: A, B, C



DFS Traversal

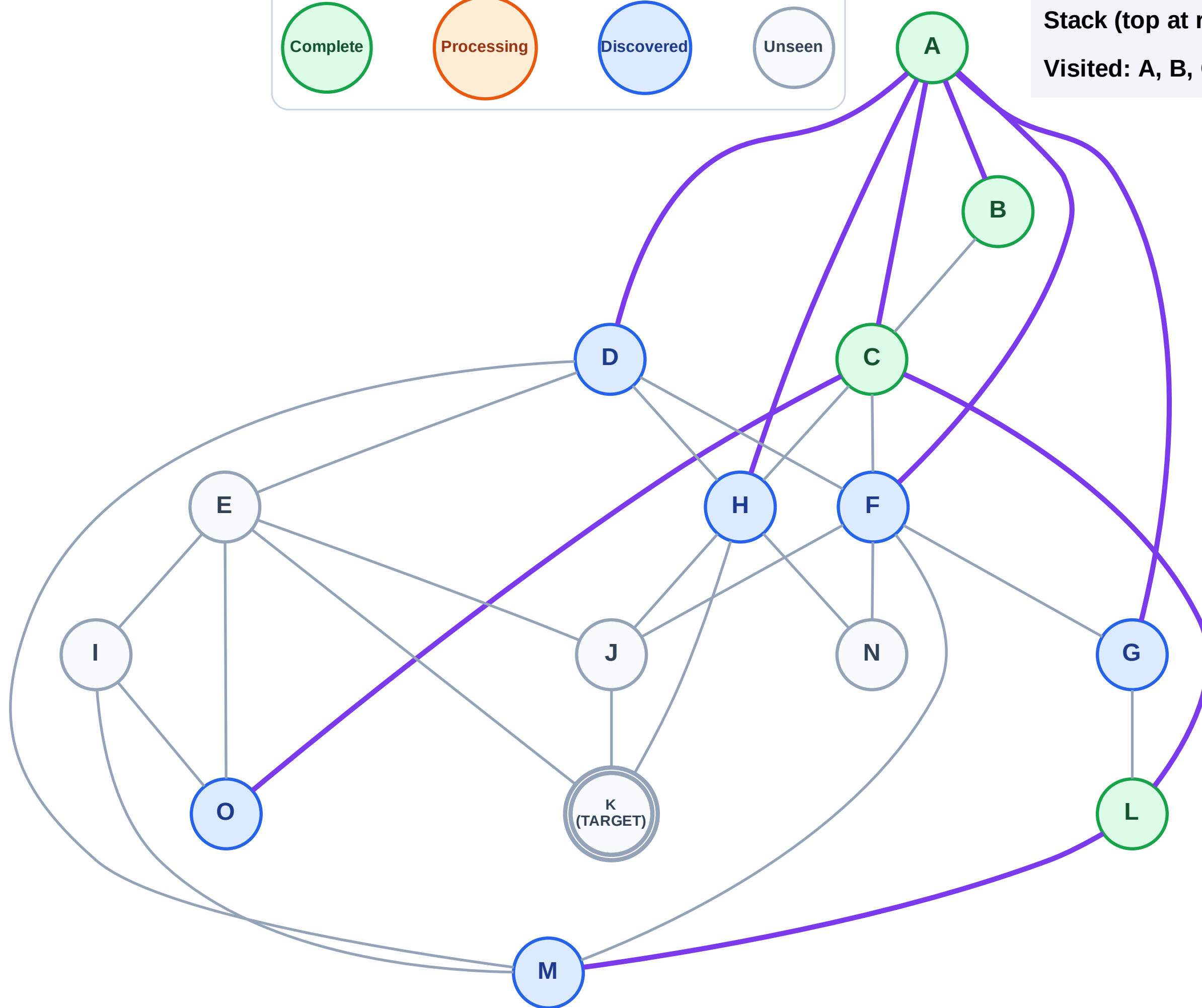
Step 9: Discover M from L

Legend



Stack (top at right): H | G | F | D | O | M

Visited: A, B, C, L



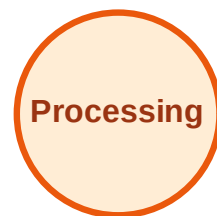
DFS Traversal

Step 10: Process node M

Legend



Complete



Processing



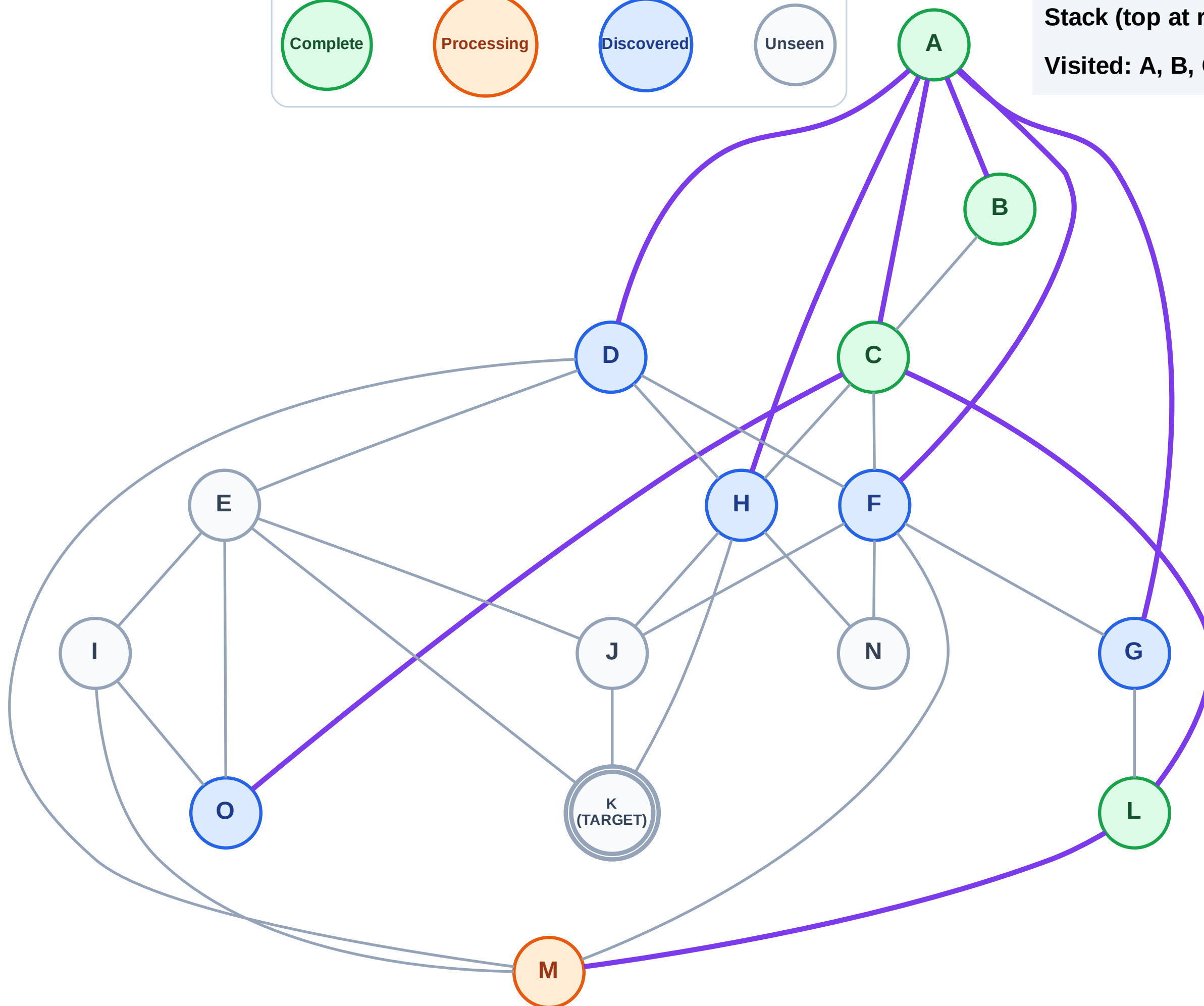
Discovered



Unseen

Stack (top at right): H | G | F | D | O

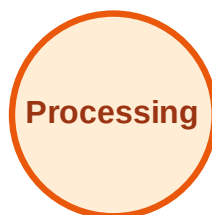
Visited: A, B, C, L



DFS Traversal

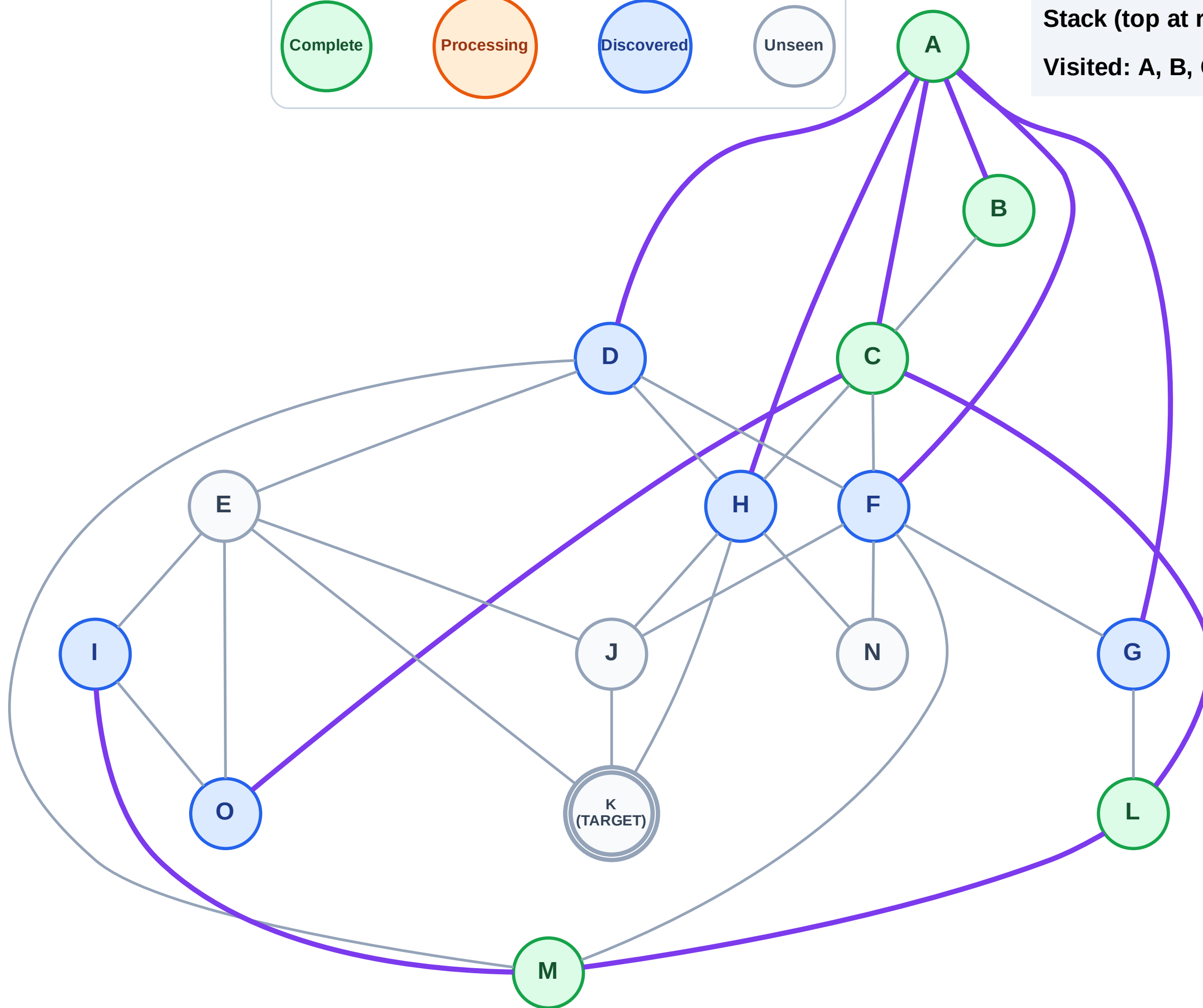
Step 11: Discover I from M

Legend



Stack (top at right): H | G | F | D | O | I

Visited: A, B, C, L, M



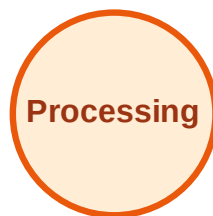
DFS Traversal

Step 12: Process node I

Legend



Complete



Processing



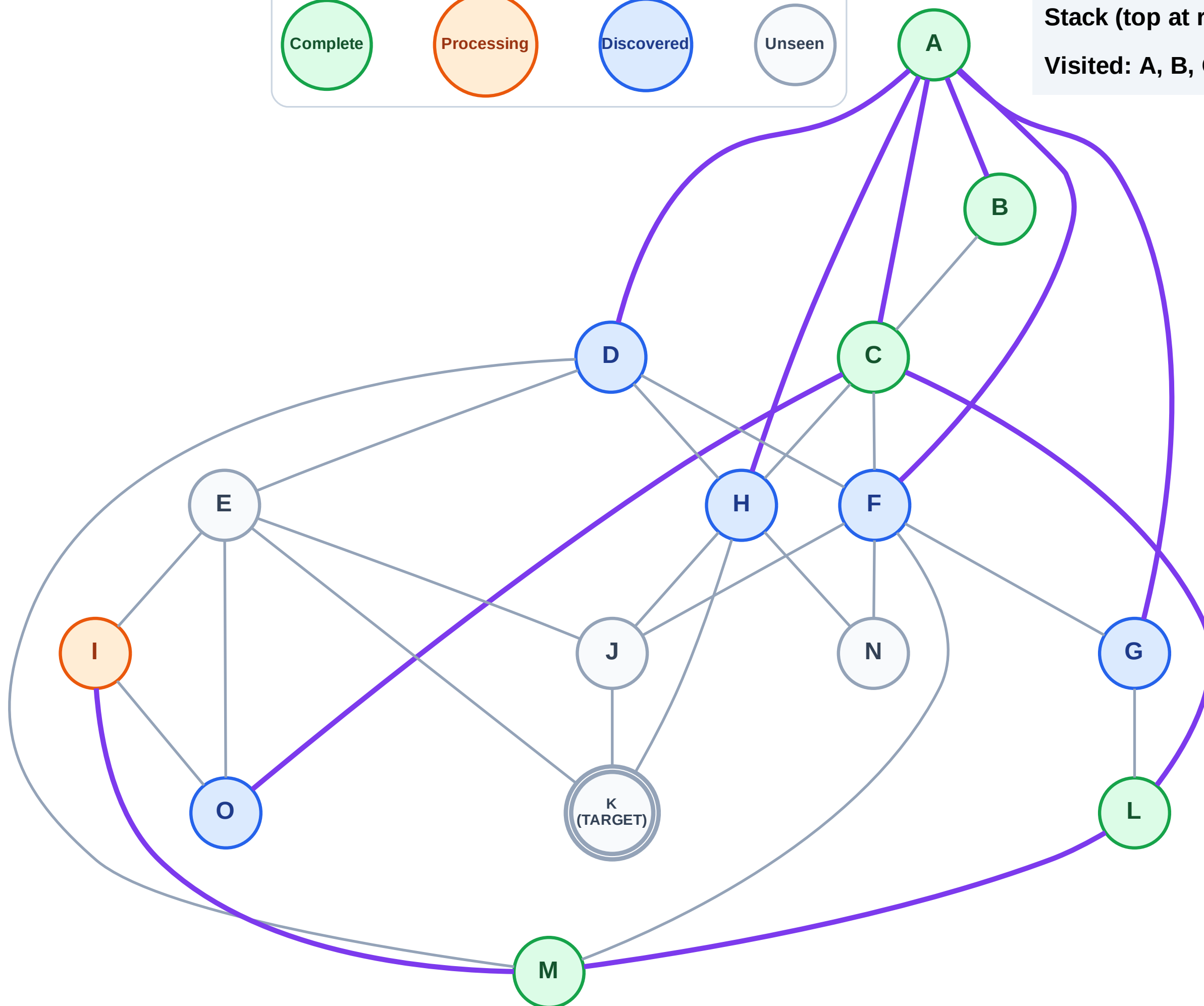
Discovered



Unseen

Stack (top at right): H | G | F | D | O

Visited: A, B, C, L, M



DFS Traversal

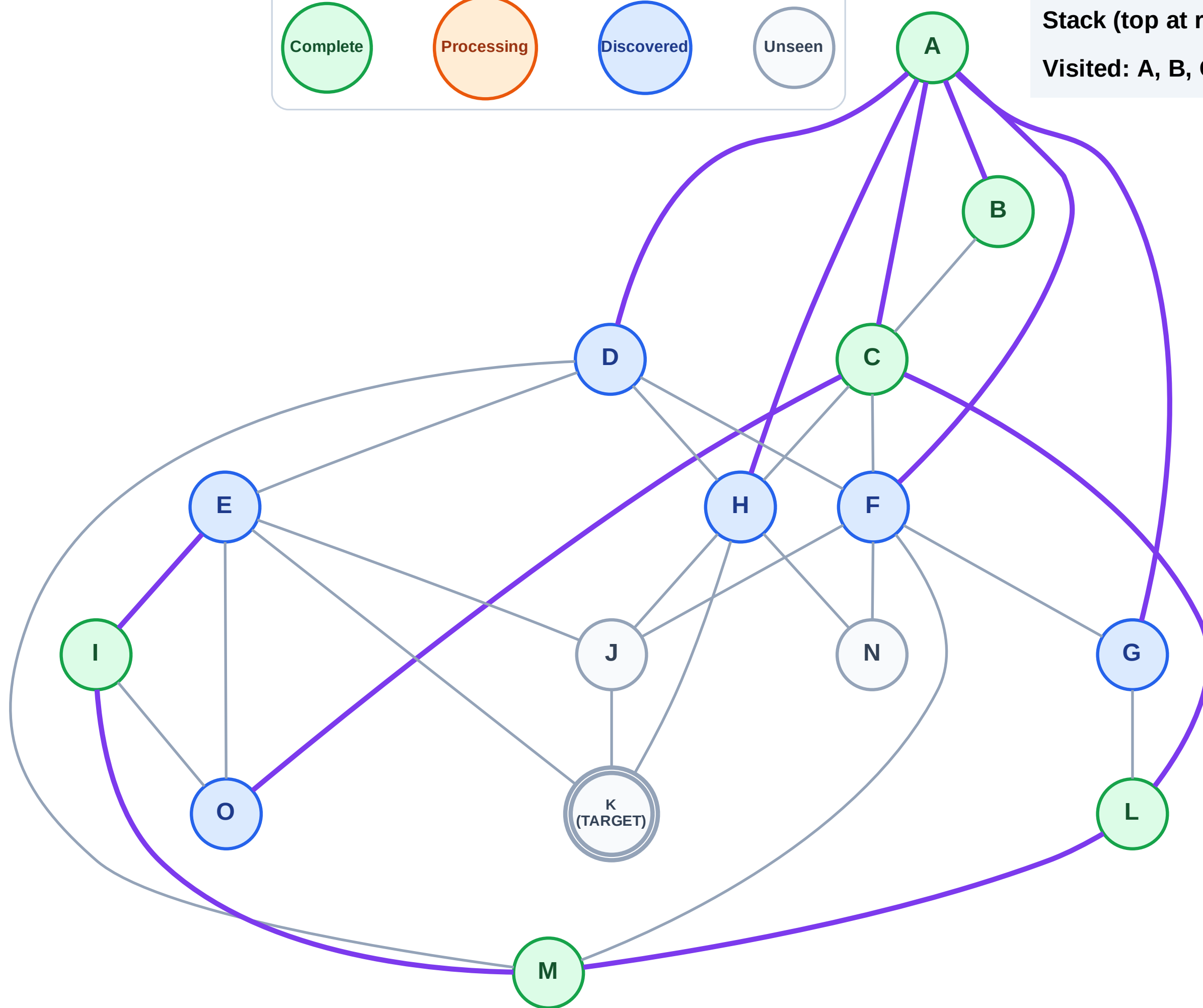
Step 13: Discover E from I

Legend



Stack (top at right): H | G | F | D | O | E

Visited: A, B, C, I, L, M



DFS Traversal

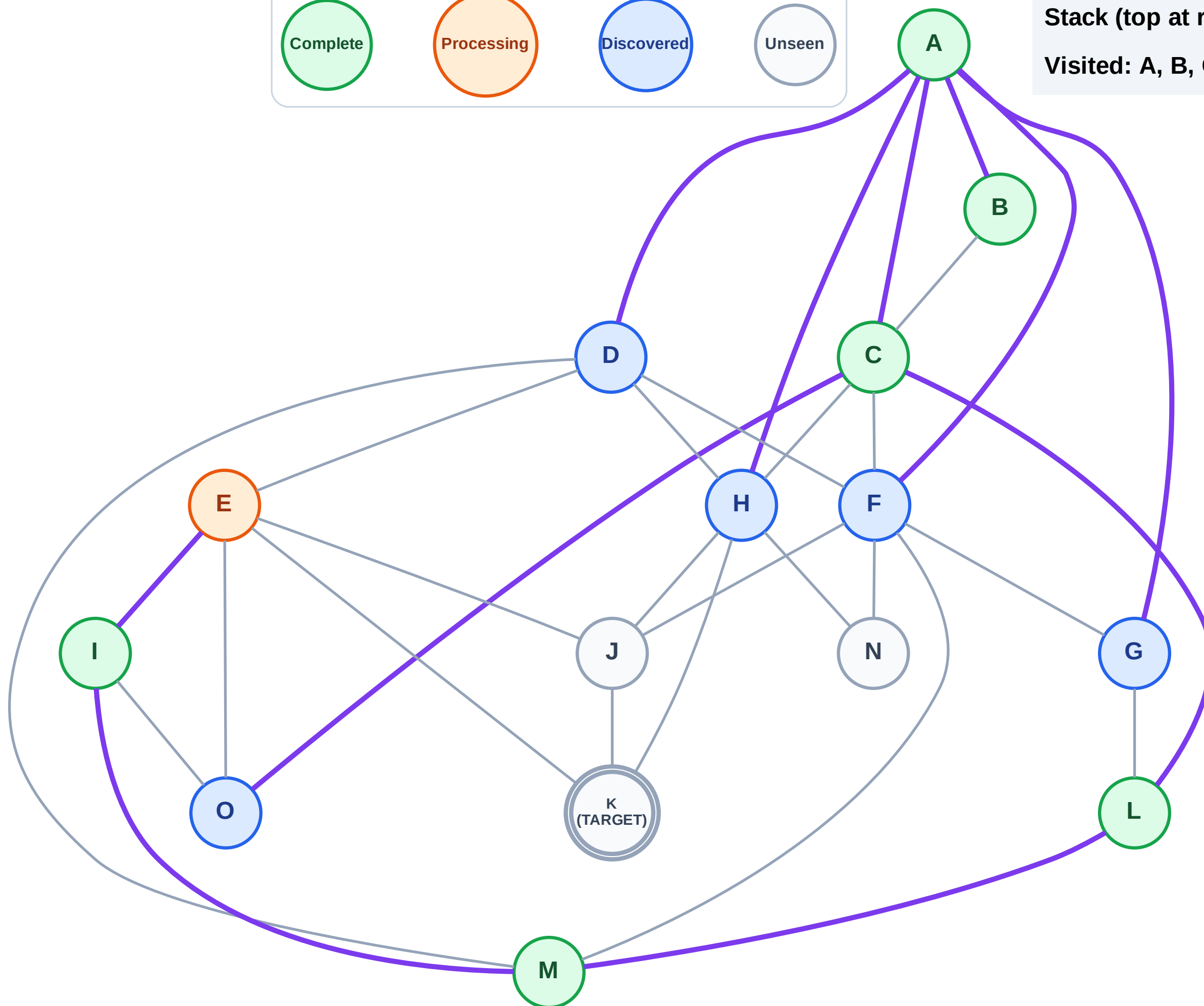
Step 14: Process node E

Legend



Stack (top at right): H | G | F | D | O

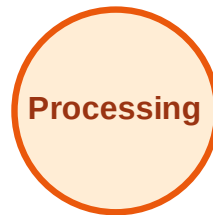
Visited: A, B, C, I, L, M



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (6 hops): A → C → L → M → I → E → K

Legend



Stack (top at right): H | G | F | D | O

Visited: A, B, C, E, I, K, L, M

